From Street Markets to Shopping Malls: The Transformation of the Service Sector*

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Job Market Paper

November 27, 2024

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Abstract: Over the development process, modern service firms like supermarkets and restaurants replace traditional micro-entrepreneurs like street vendors and food hawkers. I argue that this transformation is a central driver of service-led growth. I show empirically that modern services are more productive on the margin, and that consumer demand for them rises with income. Then I build these two features into a novel theory of the service sector. Together they imply that even a small improvement in the technology used by modern service firms sets off a cycle of amplified productivity growth. As workers transition to the modern sector, they increase aggregate income and redirect demand to modern services, which pulls in more workers and generates further growth. I estimate the model with local-level data from Brazil and show that demand effects amplified the technology-driven rise of modern services by a factor of 1.24. The resulting service transformation generated service-led growth by shifting workers into more productive jobs.

*I am indebted to my advisors, Michael Peters, Fabrizio Zilibotti, and Kevin Donovan, for their guidance throughout the many stages of this project. I thank Costas Arkolakis, Mayara Felix, Joel Flynn, Ilse Lindenlaub, Giuseppe Moscarini, Pascual Restrepo, Martin Souchier, Lauren Falcao Bergquist, Mushfiq Mobarak, Rohini Pande, Mark Rosenzweig, Nicholas Ryan, Lucas Conwell, Alvaro Cox, Bernardo Ribeiro, Tim Hersey, Sam Solomon, Lindsey Uniat, Guillermo Carranza, and numerous seminar participants at Yale, the STEG Theme 2 Workshop, and PUC-Rio for their insightful comments. I also thank the Economics Department at PUC-Rio and in particular Gustavo Gonzaga for developing the Data Zoom package to clean and analyze Brazilian survey data, an immensely useful public good. I am grateful for the financial support of the Sylff Fellowship and the STEG PhD Research Grant.

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1 Introduction

Most of the world's workers are service workers. In low- and middle-income countries, services directly surpassed agriculture as the largest sector of employment in 2012, even though the standard narrative of structural change would have these countries industrialize first.[1] The rise of services in the developing world has prompted intense interest in how to foster a productive service sector and create good service jobs (Rodrik and Sandhu, 2024). Recent evidence is optimistic: services in low-income settings show signs of strong productivity growth (Fan et al., 2023; Nayyar et al., 2021). But theories of what drives service productivity growth are scarce, especially in the developing world, where the sector primarily consists of basic consumer services like retail and food service.[2] Without a micro-founded theory of service-led growth, economists do not have a framework to extrapolate beyond the limited pool of evidence and guide policymakers on how to foster that growth.[3]

In this paper, I argue that the key to understanding service-led growth in developing countries is to recognize services as a sector in transition. I focus on the transformation that occurs *within* labor-absorbing consumer service industries like retail, hospitality, and transportation. Over the development process, *traditional* micro-entrepreneurs in these industries – the street vendors, food hawkers, and rickshaw drivers that dominate the service landscape in low-income countries – are almost totally replaced by *modern* service firms like supermarkets, restaurants, and rideshare apps. The shift from individual service providers to larger firms is not a superficial change in the face of the service sector. It represents a wholesale reorientation of services to more sophisticated forms of production.

To make this argument, I pair micro-data with a novel theory of the service sector. In the data and the theory, when workers take up modern service jobs, they generate first-order productivity gains in the style of Lewis (1954) and Song et al. (2011).[4] Due to income effects on demand, the resulting growth causes consumers to redirect expenditure towards modern services and pull even more workers into the productive modern sector, setting off a virtuous cycle of productivity growth. To identify the catalyst of this process, I use the 2000-2010 transition to modern services in all 557 local economies of Brazil to estimate the model. I find that modest improvements in the technology used by modern service firms set off an amplified cycle of service transformation. In turn, this transformation drove service-led growth by reallocating workers to more productive jobs.

[1] Source: ILO data.

[2] The long-standing view of these services was that they had little capacity for productivity growth (Baumol, 1967).

[3] Rodrik and Sandhu (2024) provide an overview of existing interventions aimed at boosting service productivity.

[4] In Song et al. (2011), productivity gains come from reallocating capital within manufacturing.

The paper begins by documenting key empirical features of services, both worldwide and in detailed micro-data on Brazilian workers and consumers. Over the development process, the composition of consumer service employment shifts from traditional self-employment without hired workers – own-account work – to wage employment at modern firms. The magnitude of this shift is large enough that essentially all of the decline in urban own-account work with development happens within consumer service industries. Panel data on Brazilian workers shows that even in poor regions where traditional services dominate, they are less productive than modern services on the margin. Workers who transition from traditional to modern services earn a wage premium, and this premium is higher in poorer places.

On the consumer side, data on household expenditures shows that rich households consume modern services more intensively than poor households. In fact, service consumption is highly segregated – a large share of households do not shop at modern service establishments at all. These patterns suggest that modern services are a luxury. Since service firms must sell to local consumers, the modern service sector is acutely sensitive to the level of local income. Modern services can only thrive when local household income is high enough to support demand.

Guided by the empirical facts, I develop a novel theory that micro-founds the transformation of services in decisions by heterogeneous consumers and workers. The model breaks from most of the literature on modernization by introducing income effects on consumer demand. Many existing models of problems like modern technology adoption assume a single homogeneous final good. This assumption is appropriate to study a sector like manufacturing in a small open economy, but it is a poor fit for services. Service providers sell a differentiated final product to local consumers. The micro-data shows that those consumers' demand for modern services is sensitive to their level of income, and that some of them do not even use modern services. I therefore draw on Lagakos (2016) to model the demand side, and assume that consumers must pay a fixed access cost to enjoy modern services, capturing the idea that these services are complemented by consumer-side investment in products like cars. The fixed cost generates the empirically relevant extensive margin of modern service consumption and implies non-homothetic demand: it is a more substantial barrier to poor consumers and therefore has more bite on service consumption when aggregate income is low.

On the labor market side, modern services use wage labor, which requires search in a frictional labor market in the vein of Diamond (1982) and Mortensen and Pissarides (1994). Traditional services have the advantage of using own-account labor, which requires no search and acts as an outside option to wage work. Crucially, search carries a cost to workers, so that some of them permanently remain in own-account work even though they are more productive as wage workers. If they do decide to search, the resulting match must pay a premium over own-account work, to compensate the worker for searching. Workers on the margin are therefore more productive in modern services than in traditional services. Inducing them to search generates aggregate income gains.

The central insight of the model is that when the economy departs from homothetic demand and frictionless labor markets, service modernization becomes a "big push" problem. Labor market frictions and income effects interact in equilibrium. The same workers who are locked out of wage work by search frictions turn to less productive traditional service work and become poor consumers with weak demand for modern services. Workers and firms do not internalize that by producing modern services together and generating a production surplus over own-account work, they increase aggregate income, reduce the bite of consumer fixed costs, and raise demand for more modern service production. When this demand externality is large enough, the economy can support multiple equilibria as in Rosenstein-Rodan (1943) and Murphy et al. (1989).

Even when the economy's equilibrium is unique, its comparative statics are amplified by the same demand externality that can create multiple equilibria, as in Buera et al. (2021). Workers do not merely compete for the same jobs; their search for wage work becomes complementary to one another as they match with firms, increase output, and foster more job creation through rising demand. Small changes in economic conditions therefore lead to large-scale reallocation towards modern services. Because workers on the margin are strictly more productive in modern wage employment, every worker who is reallocated to a modern job generates a first-order increase of aggregate productivity and welfare.

The key elements of the model – income effects on demand, and job search frictions – imply that service transformation is a powerful engine of growth. But it still requires a catalyst to set off the process of transformation. I show in the model that technological progress specific to modern service firms is a potent catalyst for service transformation. With this insight in mind, in the quantitative part of the paper, I document that modern service technology was the main underlying cause of service transformation and service-led growth within Brazil from 2000 to 2010.

In particular, I estimate the model with the ultimate goal of decomposing the causes and consequences of Brazil's service transformation: the 10 percentage point shift of Brazilian service employment towards modern services from 2000 to 2010. The key data sources are Brazilian Census data on the local allocation of employment and earnings among service workers, panel data on worker employment and earnings dynamics, and consumer expenditure data. I use the expenditure data to estimate the size of consumer income effects, as parameterized by fixed access costs. Then I pair the model with Census data to quantify search frictions, the technology level of non-service production, and the technology level of modern services for every local labor market in Brazil in 2000 and 2010.

I validate the estimated model using several untargeted moments, including the empirical response of the service sector to a natural experiment that shocked local aggregate income. For the natural experiment, I follow the work of Dix-Carneiro and Kovak (2017) and many others to estimate the local effects of exposure to import competition in the wake of Brazil's unilateral trade liberalization. Regions that were more exposed to import competition experienced a drop in the value of their industrial output, a negative shock to local income that originated outside the service sector. Consistent with the model, the income shock manifested in the service sector as a negative demand shock for modern services: it caused decreases in the modern employment share and modern wage premium among service workers. When I feed the same income shocks into the estimated model and simulate the service sector's response, the model performs well quantitatively.

In the final part of the paper, I use counterfactual simulations of the estimated model to quantify the local-level causes and consequences of service transformation within Brazil. As previewed above, I find that technological progress in modern services was the most important driver of service transformation. The key observable patterns in the data that identify modern service technology as the main catalyst of transformation are that modern service employment expanded faster than can be explained by pure income growth, and that the modern wage premium rose rather than fell. If instead transformation had been catalyzed by a reduction in search frictions, the wage premium would have declined.

The effect of modern service technology was augmented by consumer demand forces. The reinforcing effect of demand amplified the growth of output per worker in services by a factor of 1.24 compared to a counterfactual model with homothetic demand. Many models of modernization abstract away from demand-side considerations for tractability, but the quantitative results show that those models will underestimate aggregate benefits from any policy that increases local output.

Job search frictions did not drive significant service transformation from 2000 to 2010, but they are crucial to understand the consequences of reallocation in the service sector. Frictions do not only constrain the size of the modern service sector, they imply that every worker who shifts to a modern service job creates large gains to aggregate output. For example, from 2000 to 2010, real output per worker within services grew by 18% on average across Brazil. This gain would have been only half as large if modern services had not increased their share of employment. In the poorest regions of Brazil, where frictions bite especially hard, three quarters of service productivity gains came from shifting workers into more productive service jobs. These reallocation gains illustrate that frictions have a large aggregate cost, and they have direct implications for policymakers. Faced with rising modern service productivity, many developing country governments take policy stances that effectively protect traditional service jobs from competition, like weak enforcement of tax compliance and labor regulations in the traditional sector. Policy stances like these may stifle substantial growth that would follow from reallocation towards modern service jobs.

Related literature I study the productivity of the service sector in developing countries, a subject of intense interest in recent economic research (Nayyar et al., 2021; Rodrik and Sandhu, 2024). On this topic, Fan et al. (2023) measure significant growth of service productivity in Indian districts but are agnostic about its sources. In higher-income settings, Buera and Kaboski (2012) present a theory where the growth of market services stems from rising demand for more specialized and skill-intensive output. Eckert et al. (2022) argue that a decline in the price of information and communications technology (ICT) led to productivity growth in business service firms. Hsieh and Rossi-Hansberg (2023) propose a theory of an "industrial revolution in services," also precipitated by ICT, and manifested in the spatial expansion of large, modern consumer service firms.

My paper provides a bridge between the empirical literature on service productivity growth in low-income countries and theories of service-led growth in high-income countries. In my model, technological progress that favors modern firms, like ICT, also sets off service growth, but the growth mechanism is different. Rather than a mechanical effect from raising productivity in the entire service sector, gains come largely from reallocating labor out of traditional services and into more productive modern jobs, generating productivity gains on the margin in the style of Lewis (1954). This mechanism is similar to the reallocation of capital to better manufacturing firms in Song et al. (2011), or to the expansion of productive firms due to management capital (Akcigit et al., 2021; Hjort et al., 2022; Cox, 2023; Engbom et al., 2024). I focus my analysis on labor-absorbing consumer services, since skilled service professions comprise only a small fraction of employment in low-income settings.

As in studies of the "big push" like Rosenstein-Rodan (1943), Murphy et al. (1989), Matsuyama (1991), and Buera et al. (2021), the model generates a demand externality to participating in the modern sector, which can support multiple equilibria. In most of these papers, the demand externality comes from fixed costs of modern production that can only be recouped with high enough aggregate demand. In my model, it comes from workers getting richer, spending more on modern services as consumers, and drawing other workers' labor in the modern sector, so that those workers also get richer. It is a particular instance of the point made by Cooper and John (1988): that strategic complementarities can generate multiple equilibria and multipliers. In a different context, Kaplan and Menzio (2016) study multiplicity in unemployment fluctuations through a similar mechanism: differences in shopping behavior between employed and unemployed. In recent work, Walker et al. (2024) develop a model where shocks are amplified by demand effects due to input indivisibilities.

In the micro-foundations of the theory, I model own-account work as a result of frictions and subsistence concerns, in the vein of research on informality and subsistence self-employment (Ulyssea, 2010; Meghir et al., 2015; Narita, 2020; Poschke, 2023; Donovan et al., 2023; Schoar, 2010; Poschke, 2013; Breza et al., 2021; Herreño and Ocampo, 2023), and as a result of worker sorting (Gollin, 2008; Feng et al., 2024). I therefore capture two roles of own-account work: some workers turn to it out of necessity, while for others it is the best use of their labor. My micro-foundation of consumer demand includes a fixed cost of modern shopping as in Lagakos (2016) and Ramos-Menchelli and Sverdlin-Lisker (2022); Bronnenberg and Ellickson (2015) also discuss how modern retailers benefit from consumer-side investments in goods like cars and refrigerators. In my empirics, I use Bachas et al. (2023)'s classification of establishments as modern or traditional. As in that paper and Atkin et al. (2018), I find that richer consumers disproportionately shop from modern firms.

The rest of the paper is as follows. In Section 2, I document the shift to modern services in aggregate, and use micro-data to show that modern service firms have higher marginal productivity and sell to richer consumers. Section 3 lays out the model, defines the economy's equilibrium, and explores the central implications of the theory. It shows that even modest technological progress in modern services can set off substantial growth. Section 4 estimates the model at the level of local Brazilian economies and validates the estimation against several untargeted moments. In Section 5, I run counterfactual simulations of the estimated model to illustrate sources of service transformation and the magnitude of amplification and reallocation gains. Section 6 concludes.

2 Motivating Facts

Before formalizing my argument in a model, I present four facts about modern services that motivate the analysis. A coherent picture emerges from the facts: as the aggregate economy transitions to a modern service sector, individual service workers make a parallel transition out of subsistence work as micro-entrepreneurs into new careers as wage workers. When they make this transition, workers' labor becomes significantly more productive and their income rises. In turn, as richer consumers they make a consumption transition to more intensive use of modern services, so that the shift to modern services is partially self-sustaining.

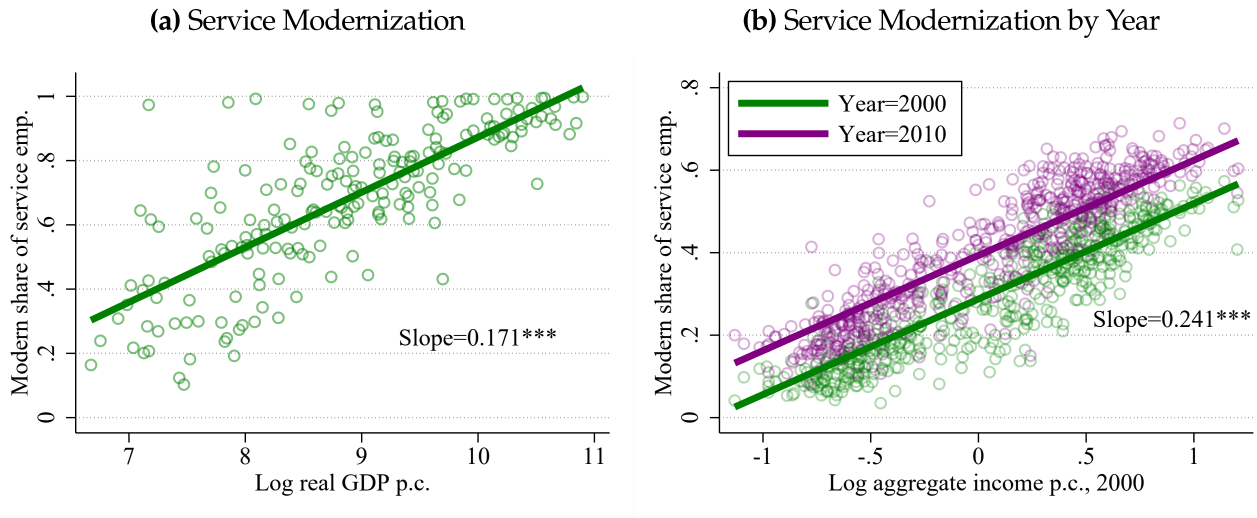
2.1 Service Transformation Along the Development Path

The modern service share is a robust indicator of economic development, both worldwide and in Brazil. Figure 1 shows how the composition of the service sector varies with aggregate income. I focus on those consumer service industries that employ a large share of workers even in low-income countries: wholesale and retail trade, hospitality and food service, transportation, and other personal services.[5] Figure 1a plots the share of these basic service workers in modern employment – defined as formal wage employment[6] – against the log of aggregate real GDP per capita for 199 country-years as observed in data from IPUMS and the Penn World Table. Figure 1b plots the same relationship among Brazilian microregions in 2000 and 2010, using local employment and income data from the decennial Census. Microregions in Brazil are a collection of municipalities that constitute a single labor market; they are analogous to commuting zones in the United States.

Across both country-years and Brazilian microregions, the modern service share rises steeply in aggregate income. The modern sector accounts for approximately 100% of service employment in the world's richest countries, but a drastically lower share in poorer places. The same patterns hold within Brazil: the modern service share is tightly linked with aggregate income in the cross-section,

[5] In Brazil, these industries account for 35% of urban employment.
[6] A worker is counted as formal if they are part of the social security system.

Figure 1: Service Modernization and Development, Worldwide and in Brazil



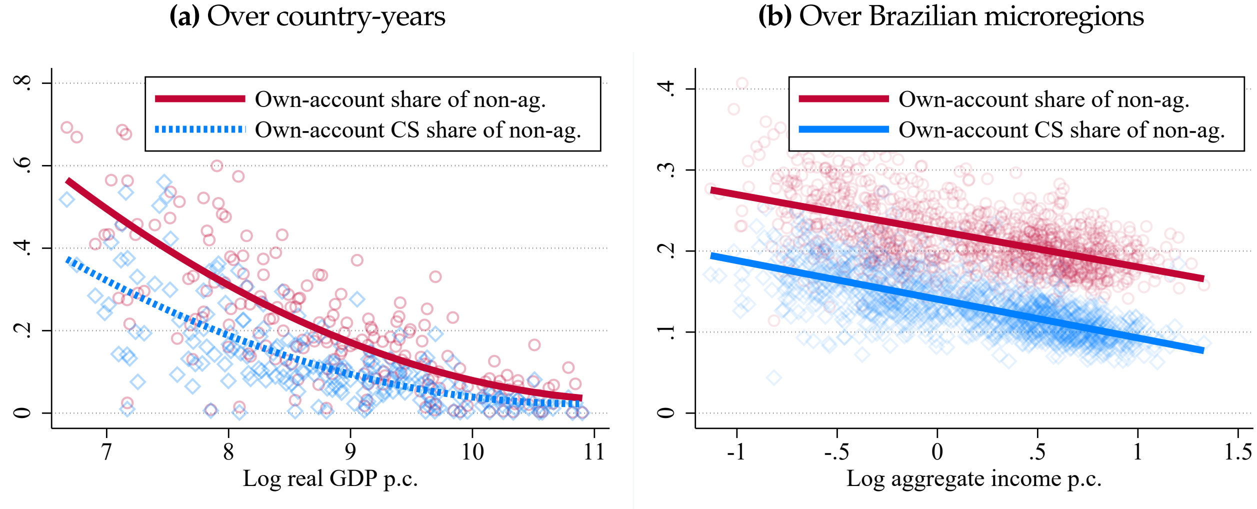
Note: Left panel: Employment data is from IPUMS data on 199 country-years spanning 76 unique countries; real GDP per capita data is from the Penn World Table for the same country-years. Right panel: data on both employment and income from the long form of Brazil's decennial Census. "Consumer services" refers to wholesale and retail trade, hospitality and food service, transportation, and other personal services. Modern employment is work for wages with pension contributions. All aggregations within a region-year use person weights provided by either IPUMS or the Brazilian Census.

and it increased by an average of 10 percentage points from 2000-2010. Notably, the rise of modern services was balanced across Brazil: poor microregions and rich microregions alike saw the same magnitude of service transformation. In the quantitative section, I document that this balanced transformation reflects across-the-board improvements in the technology used by modern service firms.

2.2 Service Transformation Drives the Decline of Own-Account Work

The modern service share is a clear development indicator. Another more widely-known development indicator is the share of self-employed or own-account workers in the economy: richer economies have a much smaller share of the labor force operating their own production unit without outside employees (Gollin, 2008; Feng et al., 2024). Figure 2 shows that these two indicators reflect the same underlying trend: the exit of service micro-entrepreneurs along the development path. Each panel of Figure 2 presents two relationships. First, the red solid line displays own-account employment as a share of total non-agricultural employment. Second, the blue dashed line displays own-account work *specifically within consumer services*, again as a share of total employment outside of agriculture. Panel 2a plots these relationships across country-years, while Panel 2b does so across microregion-years within Brazil.

Figure 2: Own-account work, services, and development



Note: Same data sources as in Figure 1: IPUMS + Penn World Table for country-years, Brazilian Census for microregion-years. Trend lines in the left panel use a quadratic fit, while in the right panel they use a linear fit.

In both cases, most of the decline in own-account work with development is accounted for by the decline in *consumer service own-account work*. Within Brazil, the decreasing share of consumer service micro-entrepreneurs accounts for the entire gradient of own-account work with respect to aggregate income; own-account work in other sectors does not decline at all with development. If high rates of own-account work in low-income settings represent subsistence self-employment, then this fact shows that subsistence labor pools specifically in the service sector.[7] As modern service businesses replace traditional micro-enterprises, the workers who operated those micro-enterprises exit subsistence labor and transition to the pool of wage labor.

2.3 Modern Services Are More Productive

The facts in Figures 1 and 2 do not necessarily imply that the shift to modern service employment causes development in and of itself. The high share of traditional and own-account service work in poor economies could simply indicate that modern services there are unproductive. If this is true, the composition of the service sector might be efficient: workers on the margin are no more productive in modern services than traditional services, so moving them into the modern sector does nothing to improve either individual income or aggregate output.

In Figure 3, I use panel data on Brazilian workers to test and reject this explanation for the

[7] For more thorough discussions and evidence of subsistence self-employment, see Schoar (2010), Poschke (2013), Breza et al. (2021), Herreño and Ocampo (2023), and Donovan et al. (2023).

high traditional share of services in poor regions. Specifically, I regress log earnings on an indicator for modern employment in a monthly panel of Brazil's urban workers, the Pesquisa Mensal de Emprego (PME).[8] I restrict the sample to service workers to test the specific hypothesis that service employment is efficiently allocated between traditional and modern services. If it is, then there should be no significant difference between modern and traditional earnings for workers on the margin between these sectors.

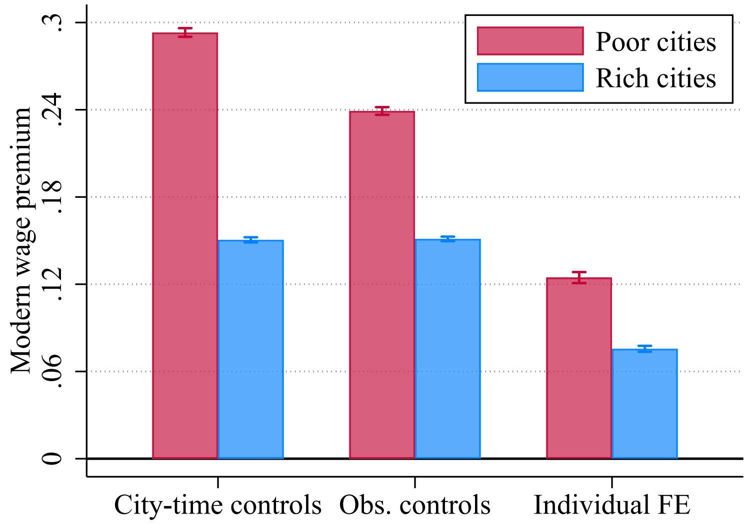
I test this hypothesis in two sub-samples: the red bars of Figure 3 present estimated sectoral earnings gaps for service workers in Brazil's underdeveloped Northeastern region, while blue bars present estimates for the more developed South and Southeast.[9] For each sub-sample, I use three different specifications of controls. First, I control only for city-month fixed effects to compare all modern and traditional service workers within that city-month. Next, I add controls for observable worker characteristics – demographics, education, and so on – to account for differences in the composition of modern and traditional workers. Finally, I use a specification controlling for individual fixed effects, so that the earnings gap is identified specifically from marginal workers who transition between modern and traditional services. I provide more detail about how I estimate the wage premium in Appendix A.2.

In all specifications and sub-samples, there is a positive and significant premium on modern work. Even after controlling for individual fixed effects, marginal workers earn 7-12% more in the modern sector – rejecting the hypothesis that their labor is equally productive across sectors. Since wage workers do not necessarily capture the full marginal product of their labor, this number represents a lower bound on the gap in marginal productivity between modern and traditional services. The gap is significantly larger in poorer regions. This fact indicates that workers in poor places face more severe barriers to joining the modern sector, and that the sizable pool of own-account service work in those places largely reflects subsistence labor. However, it also implies that shifting marginal workers into more productive modern firms represents an opportunity for significant growth.

[8] The PME was collected through monthly interviews with workers in six of Brazil's largest metropolitan areas: Recife, Salvador, Belo Horizonte, Rio de Janeiro, Sao Paulo, and Porto Alegre. Workers are interviewed in a maximum of eight months over a sixteen-month interval.

[9] In 2008, the Northeast accounted for 29% of Brazil's population but only 12% of its GDP. The Southeast accounted for 38% of population and 49% of GDP, while the South accounted for 12.5% of population and 15% of GDP.

Figure 3: Worker earning premium from modern employment, by region



Note: Data on earnings and employment status from Brazil's PME. Bars represent the coefficient from regressing the log of worker earnings on a dummy for modern employment, which refers to wage employment with pension contributions. Capped spikes represent 95% confidence intervals. "City-time controls" include fixed effects for metropolitan region interacted with month and year fixed effects; "observable controls" include the worker's age, age squared, gender, race, and education level, as well as the industry of employment; "individual FE" includes individual worker fixed effects. "Poor cities" refers to cities in Brazil's less developed Northeast, while "rich cities" refers to cities in the more developed Southeast and South.

2.4 Consumer Demand: Modern Services As Luxury

Figures 2 and 3 show that barriers to wage labor supply contribute to the low modern service share in poor economies. I now use data from Brazil's consumer expenditure survey (POF) to show that consumer demand also plays a significant role. For most purchases, the POF asks about the type of establishment where the purchase took place. I exploit this feature to classify each consumption outlay of a household as modern or traditional consumption, using the classification of consumer-facing establishments developed by Bachas et al. (2023).[10] For instance, a food hawker would count as traditional consumption, while a sit-down restaurant would count as modern consumption. I successfully classify consumption as modern or traditional for 95% of relevant household expenditures.[11]

Figure 4 highlights two notable features of modern consumption at the household level. Panel 4a presents a binned scatter plot that traces an Engel curve of the modern expenditure share against

[10] Bachas et al. (2023) construct this classification separately for a large number of countries' consumer expenditure surveys. I use the classification they developed specifically for Brazil.

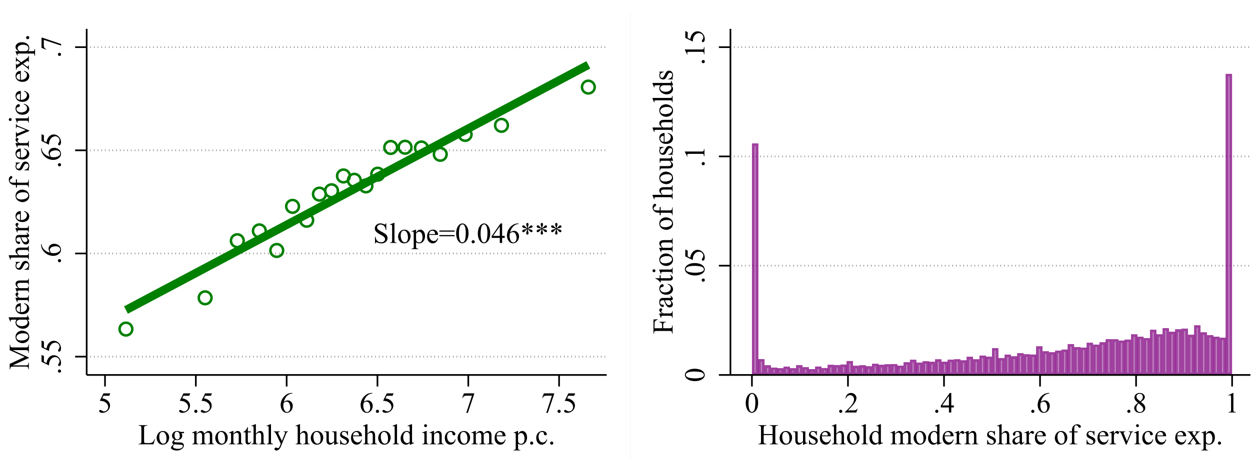
[11] This excludes certain expenditure categories, such as rent and utilities, that do not take place at an establishment; the modules of the POF that deal with these expenditures do not ask about the location of purchase.

the log of monthly household income per person. Modern services are clearly a "luxury," with their consumption following a steep income gradient. Even conditional on a battery of controls, including geography,[12] households in the richest income bin spend about 12 percentage points more of their expenditure on modern consumption than households in the poorest income bin. Appendix A.3 provides more details about the Engel curve estimation.

Panel 4b provides context to interpret the income gradient in Panel 4a. It presents a histogram of the modern share of consumption expenditure by household, revealing a striking pattern: there are large mass points near 0% and 100%. Most households consume almost exclusively from either traditional or modern establishments. I show in Appendix A.4 that this stark pattern is not explained by the presence or lack of modern establishments in the household's local economy. If it was, the location-level histogram of modern service expenditure shares would resemble the household-level histogram; Appendix A.4 shows that it does not.

Figure 4: Modern consumption expenditure among households

(a) Modern expenditure share Engel curve **(b)** Modern expenditure share histogram



Note: Data on household income and consumption establishment from Brazil's POF. I use the Bachas et al. (2023) classification of establishments to classify consumption as modern or traditional. To compute the modern expenditure share, I use household expenditures aggregated over the previous year. The binned scatter in the Engel curve includes household-level controls for number of household members, number of earners, household head's race, gender, and age by 10-year dummies. It also includes primary sampling unit (PSU) fixed effects to control for geographic location.

As households move along the Engel curve in Panel 4a, their consumption behavior shifts dramatically. Rather than gradually increasing their share of modern services, households are

[12]Specifically, I control for the household's primary sampling unit (PSU). PSUs are a small geographic unit that serve the same role in Brazilian national surveys as a Census tract in the US. There are 12,800 PSUs in Brazil, of which 4,696 were sampled for the POF, giving each PSU an average population of 15,000 in 2008. By population, a PSU is equivalent to about 1.5 US ZIP codes.

more likely to make a discrete jump in their modern service use as income rises. This observation motivates my modeling choice to introduce a fixed cost of consuming modern services, which disproportionately constrains poor households.

The facts about consumption of modern services are consistent with those found by Lagakos (2016), who documents that the county- and country-level share of modern retail – the largest consumer service industry – increases with aggregate income. Combined with the productivity gap between modern and traditional services, they imply that service transformation can be self-reinforcing. As workers move to the modern sector, their labor becomes discretely more productive and their income rises. In the product market, they become richer consumers with higher demand for modern services. This shift of demand increases the marginal revenue product of modern service labor and pulls even more workers into the modern sector, setting off a virtuous cycle of reallocation-driven growth and rising demand.

3 Theory

The previous section showed that consumer services shift to modern service firms with development, that services also host a traditional "subsistence" sector of own-account workers, that modern services pay a wage premium relative to own-account work, and that modern services are a luxury preferred by richer consumers. I now provide a novel theory of the service sector that accounts for these facts. In the model, modern service consumption carries a fixed cost that makes it a luxury, while employment in the modern sector requires workers to engage in costly search that generates a wage premium and a pool of subsistence own-account workers. Improving technology in modern services generates large employment shifts out of subsistence and into the modern sector, reinforced by the response of consumer demand. Because frictions imply an unequal marginal product of labor across sectors, this employment transition generates positive reallocation gains.

3.1 Environment

The economy in a given region r consists of a representative household. The household contains a measure L_r of members, who are heterogeneous in their ability as workers and their tastes as consumers, and owns a portfolio of firms. To produce output, workers either operate independently as self-employed entrepreneurs (own-account workers) or sell their labor to firms in a wage labor market with search and matching frictions. The household collects the production income generated by workers and firms and distributes it equally among consumers, who make independent consumption decisions. I denote the representative level of consumption expenditure by e_r. Time is continuous and the household has discount rate ρ. Because there is no aggregate uncertainty, all of the household's workers and firms simply maximize the expected present discounted value of their income. And because there is no capital or other investment good, net savings are zero and the household's total expenditure always equals total income. The remainder of the model exposition will drop regional subscripts r until the quantitative section, which uses local-level data from each microregion of Brazil. I focus on the decentralized stationary equilibrium of the economy, where the real interest rate equals ρ.

3.1.1 Consumer Preferences

To capture the empirically relevant extensive margin of modern service consumption, I model consumers as facing a static discrete choice between a traditional or modern consumption bundle.

Key assumption: Fixed cost of modern consumption A key empirical feature of consumer choice is non-homothetic demand: modern services are a luxury good. I therefore introduce income effects by assuming that modern consumption requires consumers to pay a fixed cost. The fixed cost captures the idea that the modern service sector requires a complementary input from consumers. This input includes transportation and storage technologies, like the crucial role of cars emphasized by Lagakos (2016)[13] or fuel in Ramos-Menchelli and Sverdlin-Lisker (2022); Bronnenberg and Ellickson (2015) highlight refrigerators as another consumer-side investment that facilitates modern services. Beyond the many examples of consumer inputs into consumption, the fixed cost is an empirically grounded way of capturing consumer income effects with a single parameter.

Consumers face a static choice between a modern consumption bundle with price $P_{\overline{M}}$ and a traditional bundle with price $P_{\overline{T}}$. They allocate their entire expenditure e to whichever bundle they choose. If they choose the modern bundle, not all of their expenditure translates directly into consumption. They must devote a fixed number χ_c of units of the modern bundle to covering the fixed cost of consumption.

[13] In Lagakos (2016), cars are not strictly necessary to consume modern retail, but they reduce the required time input of consumers.

Each consumer j has idiosyncratic utility β_k^j from consuming bundle k, so that the indirect utility of each consumption bundle for consumer j is as follows:

$$V_M^j = \beta_M^j \left(\frac{e}{P_{\overline{M}}} - \chi_c \right), \quad V_T^j = \beta_T^j \frac{e}{P_{\overline{T}}} \tag{1}$$

Notice that the utility from consuming the modern bundle V_M^j must subtract the fixed cost χ_c from real modern purchases $\frac{e}{P_{\overline{M}}}$. The consumer does not enjoy any consumption utility from the χ_c units of the modern composite that she must buy to meet the fixed cost.

Each consumer chooses her consumption bundle based on the realization of her taste shocks β_k^j. An endogenous share of consumers, which I denote ϑ_M, consumes the modern bundle if their relative taste shock for it is sufficiently high. ϑ_M depends on the endogenous prices $P_{\overline{M}}$ and $P_{\overline{T}}$, nominal expenditure e, and the exogenous iceberg fixed cost χ_c:

$$\vartheta_M(P_{\overline{M}}, P_{\overline{T}}, e | \kappa_c) = \mathbf{Pr}\left(V_M^j \geq V_T^j\right) = \mathbf{Pr}\left(\frac{\beta_M^j}{\beta_T^j} \geq \frac{P_{\overline{M}} \Big/ \left(1 - \frac{\chi_c}{e/P_{\overline{M}}}\right)}{P_{\overline{T}}} \right) \tag{2}$$

ϑ_M therefore decreases in the relative price of the modern bundle $\frac{P_{\overline{M}}}{P_{\overline{T}}}$ and the fixed cost χ_c. When $\chi_c > 0$, it also increases with respect to the consumer's purchasing power over modern consumption $\frac{e}{P_{\overline{M}}}$. The fixed cost χ_c implies that consumer demand is non-homothetic and subject to income effects.

These effects become even clearer after imposing structure on the taste shocks β_k^j. I assume going forward that the preference shock for bundle k is distributed Fréchet with shape parameter $\xi - 1$ and location parameter β_k, where $\beta_M + \beta_T = 1$. In this case, the choice probability ϑ_M resembles CES demand:

$$\frac{\vartheta_M}{1 - \vartheta_M} = \begin{cases} \frac{\beta_M}{1 - \beta_M} \left(\frac{P_{\overline{M}}}{P_{\overline{T}}} \right)^{1-\xi} \left(\frac{\frac{e}{P_{\overline{M}}} - \chi_c}{\frac{e}{P_{\overline{M}}}} \right)^{\xi - 1} & \text{if } \frac{e}{P_{\overline{M}}} > \chi_c \\ 0 \quad \text{otherwise} \end{cases} \tag{3}$$

When $\vartheta_M > 0$, the expression for it is CES with an adjustment for real modern purchasing power $\frac{e}{P_{\overline{M}}}$ relative to the fixed cost χ_c. Holding relative prices $\frac{P_{\overline{M}}}{P_{\overline{T}}}$ fixed, the modern sector claims a greater share of household expenditure as the household's purchasing power rises. If purchasing power does not exceed χ_c, then no consumers shop from the modern sector at all.

3.1.2 Production

I now lay out production functions in the economy, which provide a link between consumer choice and labor market outcomes like employment and wages. Production functions are simple, to leave the model tractable while having rich predictions about consumer demand and labor supply.

Production consists of intermediate goods and services and final consumption bundles. Final bundles are each a Cobb-Douglas composite of intermediate goods X (with intensity $1-\phi$) and either intermediate modern services Y_M or intermediate traditional services Y_T (with intensity ϕ). The service content therefore distinguishes the bundles. Intermediate goods and services are all produced using efficiency units of labor as the only factors of production. Traditional services are produced by autonomous own-account workers with undifferentiated output. Modern services and goods are aggregated from an endogenous measure of equally productive and differentiated firms, using a CES aggregator without love-of-variety. Production functions are as follows:

Final consumption bundles:

$$\text{Traditional bundle:} \quad B_{\overline{T}} = \frac{Y_T^{\phi} X^{1-\phi}}{\phi^{\phi}(1-\phi)^{1-\phi}} \tag{4}$$

$$\text{Modern bundle:} \quad B_{\overline{M}} = \frac{Y_M^{\phi} X^{1-\phi}}{\phi^{\phi}(1-\phi)^{1-\phi}}$$

Intermediates:

$$\text{Traditional services:} \quad Y_T = \psi_T H_T$$

$$\text{Modern services:} \quad Y_M = \left(\int_0^{N_M} \left(\frac{1}{N_M} \right)^{\frac{1}{\sigma_M}} \left(a_M h_{M,n}^P \right)^{\frac{\sigma_M - 1}{\sigma_M}} dn \right)^{\frac{\sigma_M}{\sigma_M - 1}} \tag{5}$$

$$\text{Intermediate goods:} \quad X = \left(\int_0^{N_X} \left(\frac{1}{N_X} \right)^{\frac{1}{\sigma_X}} \left(\psi_X h_{X,n}^P \right)^{\frac{\sigma_X - 1}{\sigma_X}} dn \right)^{\frac{\sigma_X}{\sigma_X - 1}},$$

where $h_{k,n}^P$ is the measure of production labor used by firm n in sector $k \in \{M,X\}$. To operate, firms in sector k need f_k units of overhead labor. Solving the firms' free entry problem generates aggregate production of modern services $Y_M = A_M H_M$ and goods $X = \Psi_X H_X$, where $A_M = \frac{\sigma_M - 1}{\sigma_M} a_M, \Psi_X = \frac{\sigma_X - 1}{\sigma_X} \psi_X$. H_T, H_M, H_X respectively denote aggregate efficiency units supplied to each intermediate sector. In each sector $k \in \{M,X\}$, an endogenous measure of firms $N_k = \frac{1}{\sigma_k} \frac{H_k}{f_k}$

operates. I denote the respective output prices of intermediates as q_T,q_M,q_X.

3.1.3 Labor supply

There are two factors of production: efficiency units of own-account labor and wage labor. Traditional services use own-account labor for production, while goods and modern services use wage labor. Each type of labor is supplied by workers, who sort between own-account work in the traditional sector without search, or own-account work with costly search. If search is successful, the worker supplies her wage labor to production of modern services or goods for the duration of her match to the employer. This model of the labor market resembles the one in Feng et al. (2024), but without an unemployment option. It respects the key facts about employment in consumer services: that own-account work in traditional services represents a "subsistence" option for many workers, and that employment in modern service firms pays a premium over own-account work among workers who make that transition. Since I focus on the steady state of the economy, I omit time derivatives from the relevant Bellman equations.

Worker heterogeneity To illustrate the economics of the labor market, I begin by assuming that workers are heterogeneous in one dimension: own-account ability z. Specifically, a given worker i can supply either 1 efficiency unit of wage labor or z_i efficiency units of own-account labor; z_i captures her comparative advantage in own-account work. Ability is permanent and not subject to shocks. In the quantitative section, I discuss a tractable way to additionally make workers heterogeneous in their ability as wage workers, to better capture earnings patterns.

Own-account work without search If the worker chooses not to search, she uses her ability z_i to produce in the own-account sector permanently. I denote this employment state O_p. The HJB equation for it is straightforward:

$$\text{Own-account without search:} \quad \rho O_p(z) = q_T \psi_T z = w_T z, \tag{6}$$

where q_T is the output price of traditional services, so that $w_T = q_T \psi_T$ is the skill price of traditional own-account work. ψ_T is the aggregate technology level of traditional services.

Search in the frictional labor market The worker can instead choose own-account work with search, denoted O_s. If she searchers, at endogenous rate $\lambda(\theta(z))$, she matches with an employer and enters wage work W. While matched, she earns a bargained flow wage $w_e(z)$ until the match separates at exogenous rate δ. Upon separation, the worker returns to own-account work with search. The worker's HJB equations if she searches are:

$$\begin{aligned} \text{Own-account with search:} \quad & \rho O_s(z) = w_T z + \lambda(\theta(z))(W(z) - O_s(z)) \\ \text{Wage work:} \quad & \rho W(z) = w_e(z) + \delta(O_s(z) - W(z)) \end{aligned} \tag{7}$$

Employers and firms There is a single employment sector for wage labor that pays all type-z workers the same wage $w_e(z)$. Employers in this sector act as intermediaries between workers and producers: they match with workers in the frictional labor market and then sell their labor to producers of modern services and goods. When intermediaries sell to producers, they act as price-takers selling labor at the "firm wage" w_f. This assumption about the labor market ensures that the marginal revenue product of labor is equal between modern services and goods when both sectors operate: $w_f = q_M A_M = q_X \Psi_X$. As in Moscarini and Postel-Vinay (2023), it also avoids a complicated problem where firms have pricing power in both the labor market and the product market.

The employer's vacancy-posting problem Employers hire by posting vacancies, which carry a flow cost of κ_v units of wage labor. Labor markets are segmented by z: employers can direct their vacancies to attract type-z workers specifically. This produces z-specific market tightness $\theta(z)$. The matching function has constant returns to scale, so that workers match with employers at rate $\lambda(\theta(z))$ and vacancies are filled at rate $\frac{\lambda(\theta(z))}{\theta(z)}$. A filled job produces revenue w_f and the employer pays the worker the "employee wage" $w_e(z)$, which is set through Nash bargaining over the match surplus. Workers have bargaining weight η and employers have weight $1-\eta$. The resulting HJB equations for the employer are:

$$\begin{aligned} \text{Filled job with type } z: \quad & \rho J(z) = w_f - w_e(z) - \delta J(z) \\ \text{Vacancy for type } z: \quad & \rho V(z) = -\kappa_v w_f + \frac{\lambda(\theta(z))}{\theta(z)} J(z) \end{aligned} \tag{8}$$

There is free entry into vacancies, so that in equilibrium $V(z) \leq 0$ for all z.

Solving the z-specific labor market Having specified all value functions and the employer's vacancy-posting problem, it is possible to solve for z-specific match surplus $M(z)$, market tightness $\theta(z)$, and wages $w_e(z)$ for any z such that employers post vacancies. These satisfy the following equations:

$$\begin{aligned} \text{Match surplus:} \quad & M(z) = \frac{w_f - w_T z}{\rho + \delta + \eta\lambda(\theta(z))}, \\ & W(z) - O_s(z) = \eta M(z),\ J(z) = (1-\eta)M(z) \\ \text{Market tightness:} \quad & \kappa_v \theta(z) = \frac{1 - \frac{w_T}{w_f} z}{\rho + \delta + \eta\lambda(\theta(z))} \\ \text{Employee wage:} \quad & w_e(z) = \eta w_f + (1-\eta) w_T z + \eta \kappa_v w_f \theta(z) \end{aligned} \tag{9}$$

Notice that this generates a z-specific wage premium over traditional own-account work:

$$w_e(z) - w_T z = \eta(w_f - w_T z + \kappa_v w_f \theta(z)) \tag{10}$$

The labor market to this point follows a completely standard vacancy-posting model, with markets segmented by workers' outside option. The flow earnings from this outside option are endogenous since they depend on the equilibrium skill price w_T. This implies that the match surplus and market tightness respond to shifts of aggregate demand.

Key assumption: Costly worker search I now make a departure from the standard vacancy-posting model. In order to enter the frictional labor market and search for wage employment, workers must pay an up-front search cost. They must hire $\frac{\kappa_w}{\rho}$ units of wage labor to pay for initial job search services, leading to a search entry cost of $\frac{\kappa_w}{\rho} w_f$. Note that the PDV of this cost to workers is equivalent to paying a flow cost $\kappa_w w_f$ in perpetuity. Job search services capture the permanent investments workers make in order to effectively search for wage work, including job application workshops as in Abebe et al. (2021), and credentialing as in Alfonsi et al. (2020). Evidence from these interventions shows that up-front costly investments in worker search have long-run positive effects on employment and earnings. I provide a thorough discussion about this assumption on worker search costs, as well as alternative assumptions, in Appendix B.5.

The worker's problem Search carries a fixed cost to workers but brings heterogeneous returns due to differences in z. The worker searches only if her value of adding search to own-account work, $O_s(z_i)-O_p(z_i)$, exceeds the search entry cost $\frac{\kappa_w}{\rho}w_f$. The worker's problem of whether or not to search therefore generates a Roy (1951) model where workers select into search based on the returns relative to the up-front cost. Specifically, their binary decision of whether or not to search for wage work $s(z_i)$ follows a threshold rule with respect to z_i:

$$s(z_i)=\begin{cases}1 & \text{if } z_i\leq z_w\\ 0 & \text{if } z_i>z_w,\end{cases} \tag{11}$$

$$\text{where } z_w=\frac{w_f}{w_T}\left(1-\frac{\rho+\delta+\eta\lambda(\theta(z_w))}{\eta\lambda(\theta(z_w))}\kappa_w\right) \tag{12}$$

In the absence of worker search costs, $\kappa_w=0$, any worker who is more productive as a wage worker than an own-account worker (i.e. with $z\leq\frac{w_f}{w_T}$) searches. Worker search costs $\kappa_w>0$ generate a deviation from this sorting rule. Frictions in the form of search costs therefore limit the pool of labor available for modern service production, reducing consumer purchasing power. They also generate subsistence labor: the pool of workers who are more productive in the modern sector, but who continue producing traditional services as own-account workers due to frictions. And they imply that those workers who do participate in the modern sector must be compensated by a large wage premium, to make up for the cost of search.

Most importantly, the marginal searchers z_w are strictly more productive in the modern sector: $w_f=q_MA_M>w_Tz_w=q_T\psi_Tz_w$. Moving workers out of subsistence and into the modern sector therefore generates a large Lewisian gain.

3.1.4 Market Clearing

The Cobb-Douglas setup of final production simplifies the market-clearing conditions of the model. In particular, it implies that expenditure on intermediates is proportional to expenditure on final consumption bundles:

$$q_MY_M=\phi P_{\overline{M}}B_{\overline{M}},\quad q_TY_T=\phi P_{\overline{T}}B_{\overline{T}},\quad q_XX=(1-\phi)(P_{\overline{M}}B_{\overline{M}}+P_{\overline{T}}B_{\overline{T}}) \tag{13}$$

In turn, the modern consumption bundle takes a share ϑ_M of final expenditure, so that $P_{\overline{M}} B_{\overline{M}} = \vartheta_M eL$ and $P_{\overline{T}} B_{\overline{T}} = (1-\vartheta_M)eL$, resulting in the following conditions linking consumer demand ϑ_M to demand for sector-specific efficiency units of labor:

$$
\begin{aligned}
\phi \vartheta_M eL &= q_M Y_M = w_f H_M \\
\phi(1-\vartheta_M)eL &= q_T Y_T = w_T H_T \\
(1-\phi)eL &= q_X X = w_f H_X,
\end{aligned}
\tag{14}
$$

where the second equality in each line comes from writing intermediate production in terms of efficiency units.

Meanwhile, the steady-state supply of labor efficiency units is implied by the frictional labor market laid out above. In particular, it depends on z_w as follows:

$$
H_M + H_X = H_w = \int_0^{z_w} \frac{\lambda(\theta(z)) - \kappa_v \theta(z)}{\lambda(\theta(z)) + \delta} f_Z(z) dz \tag{15}
$$

$$
H_T = \int_0^{z_w} \frac{\delta}{\lambda(\theta(z)) + \delta} z f_Z(z) dz + \int_{z_w}^{\infty} z f_Z(z) dz, \tag{16}
$$

where $f_Z(z)$ is the density function of the distribution of z over workers, and market tightness $\theta(z)$ is as in (9). I derive the steady-state labor supply in Appendix B.2.

Finally, total expenditure must equal total revenues from production, in turn equal to the total revenue product of labor:

$$
eL = q_M A_M H_M + q_X \Psi_X H_X + q_T \psi_T H_T = w_f H_w + w_T H_T, \tag{17}
$$

3.2 Equilibrium: Consumer and Worker Choice

Having laid out consumer preferences, production, and the labor market, and having closed the model by linking skill prices to final prices, it is now possible to define the equilibrium of the economy.

Definition 1. *A steady state equilibrium is an allocation of aggregate sectoral labor efficiency units* (H_M, H_T, H_X)*, final prices* $(P_{\overline{M}}, P_{\overline{T}})$*, skill prices* (w_f, w_T)*, expenditure level* e*, labor market tightness* $\theta(z)$*, consumer expenditure share* ϑ_M*, and worker sorting threshold* z_w*, such that:*

1. *ϑ_M is consistent with optimal consumer choice (3) given $(P_{\overline{M}},P_{\overline{T}})$ and e.*

2. *z_w is consistent with optimal worker sorting (12) given (w_f,w_T) and $\theta(z)$.*

3. *All markets for final bundles, intermediate goods and services, and efficiency units of labor clear, and all revenues are used for consumer expenditure, as summarized by Equations (13), (14), (16), and (17).*

Despite the rich structure of the economy, it turns out that equilibrium hinges on just two endogenous variables: consumers' modern expenditure share ϑ_M and workers' sorting threshold into search z_w. Production and vacancy-posting decisions follow mechanically from these key variables, and hence so does the equilibrium allocation (H_M,H_T,H_X). This fact is formalized in the following lemma:

Lemma 1. *All allocations and relative prices in the economy can be written functions of z_w, ϑ_M, and exogenous parameters. For example, one can write $\frac{w_f}{w_o}(z_w,\vartheta_M)$, $\frac{P_{\overline{M}}}{P_{\overline{T}}}(z_w,\vartheta_M)$, and $H_M(z_w,\vartheta_M)$.*

As a result, the worker search margin z_w is a function only of ϑ_M and exogenous parameters, and the modern consumption share ϑ_M is a function only of z_w and exogenous parameters:

$$z_w = z_w^*(\vartheta_M), \quad \vartheta_M = \vartheta_M^*(z_w) \tag{18}$$

Therefore the equilibrium can be characterized by a tuple (z_w,ϑ_M), where z_w and ϑ_M are mutually consistent with workers' and consumers' optimal choices. For details and a proof, see Appendix B.3.

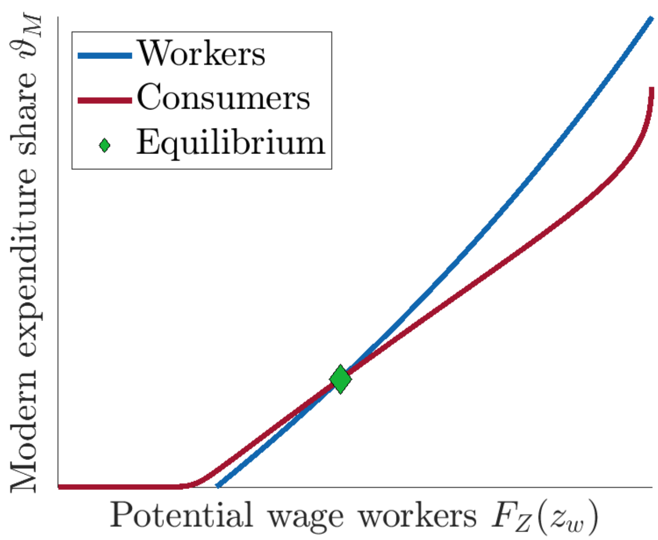
All relative prices are themselves functions of only z_w and ϑ_M.[14] These endogenous variables move prices in a natural way. More worker search z_w means that modern wage labor is more abundant, which pushes down the relative skill price $\frac{w_f}{w_o}$ and relative final price $\frac{P_{\overline{M}}}{P_{\overline{T}}}$ when demand for modern consumption – captured by ϑ_M – is held fixed. ϑ_M has the opposite effect: when the supply of wage work searchers – captured by z_w – is held fixed, raising desired expenditure on the modern sector pushes up relative wages and relative modern prices. The consumer choice ϑ_M also depends on real modern purchasing power $\frac{e}{P_{\overline{M}}}$; this too can be written as a function of just ϑ_M and z_w. In particular, z_w raises real purchasing power, because shifting workers into search makes modern wage labor more plentiful and leads to long-run income gains.

To visualize the equilibrium, I plot optimal worker sorting $z_w^*(\vartheta_M)$ and consumer choice

[14] Of course, consistent with market clearing conditions.

$\vartheta^*_M(z_w)$ in Figure 5. The blue line corresponds to worker sorting: the share $F_Z(z_w)$ of workers (on the horizontal axis) who sort into searching based on the choices ϑ_M of consumers. It is upward-sloping because more expenditure on the modern sector raises the relative skill price of wage work. Meanwhile, the red line corresponds to consumer choice: the share ϑ_M of consumers who consume from the modern sector based on worker sorting z_w. If z_w is too low, ϑ_M is zero: so few workers search that the economy is too poor to pay the fixed cost of modern shopping. Once z_w is sufficiently high, $\vartheta^*_M(z_w)$ is upward-sloping. This is for two reasons. First, z_w lowers the relative price of modern consumption. Second, z_w increases aggregate real income. This second effect is the income effect of z_w on demand. That both $z^*_w(\vartheta_M)$ and $\vartheta^*_M(z_w)$ are upward-sloping shows that workers' and consumers' choices to participate in the modern sector are strategic complements.

Figure 5: Equilibrium in $(F_Z(z_w),\vartheta_M)$ space



At the point in Figure 5 where worker sorting and consumer choice intersect, workers and consumers are responding optimally to one another. This point therefore represents the steady state equilibrium of the economy. Despite the rich elements of the theory – including non-homothetic demand, search and matching frictions, worker heterogeneity, and sorting – to solve for equilibrium, it is sufficient to find a fixed point of (z_w,ϑ_M). This property of the model contributes to its computational tractability, simplifying both estimation and subsequent simulations.[15]

[15] In the quantitative section, the market tightness $\theta(z)$ has an analytic solution, so that only z_w and ϑ_M require numerical methods to solve for.

3.2.1 Welfare and Demand Externality

Before discussing growth in this economy, it is useful to first define the welfare-relevant notion of aggregate output: real consumption.

Definition 2. *Under Fréchet taste shocks over final consumption bundles, real stationary consumption* $\mathbf{C}$ *is as follows:*

$$\mathbf{C}=\left(\beta_M^{\frac{1}{\hat{\xi}}}\left(\frac{Y_M}{(1+\mathcal{T}_c)^{\frac{1}{\phi}}}\right)^{\frac{\hat{\xi}-1}{\hat{\xi}}}+(1-\beta_M)^{\frac{1}{\hat{\xi}}}(Y_T)^{\frac{\hat{\xi}-1}{\hat{\xi}}}\right)^{\frac{\hat{\xi}}{\hat{\xi}-1}\phi}\frac{X^{1-\phi}}{\phi^{\phi}(1-\phi)^{1-\phi}}, \tag{19}$$

where $\hat{\xi}-1=\phi(\xi-1)$*, and* $\mathcal{T}_c=\frac{\chi_c}{\frac{e}{P_{\overline{M}}}-\chi_c}$ *is an implicit tax on modern consumption due to fixed costs.*

The consumption aggregate is a Cobb-Douglas composite of real service consumption – which is a CES aggregate over modern and traditional services – and goods consumption. Real service consumption must adjust for the fixed cost χ_c through the term $(1+\mathcal{T}_c)^{\frac{1}{\phi}}$. $\mathcal{T}_c$ is the implicit tax on modern consumption. Consumers experience modern consumption as if it were subject to a tax $\mathcal{T}_c$, and ϑ_M behaves accordingly:

$$\frac{\vartheta_M}{1-\vartheta_M}=\frac{\beta_M}{1-\beta_M}\left(\frac{(1+\mathcal{T}_c)P_{\overline{M}}}{P_{\overline{T}}}\right)^{1-\xi} \tag{20}$$

Welfare-relevant consumption has to take $\mathcal{T}_c$ into account. From the perspective of firms, $\mathcal{T}_c$ is like an endogenous "output wedge" in the vein of Hsieh and Klenow (2009), faced by all modern firms.

Demand Externality The endogenous nature of the implicit tax implies that any decentralized equilibrium of the model is generically inefficient, even when the Hosios (1990) condition holds in the labor market. Indeed, it is inefficient from the perspective of any social planner who wants to maximize consumer welfare and takes the marginal rate of transformation between modern and traditional services as given. To see this, it is easiest to compare the relative private value of modern service output to its relative social value. The relative private value is simply reflected by the relative service price $\frac{q_M}{q_T}$. The relative social value can be found by taking partial derivatives of consumer welfare $\mathbf{C}$:

$$\begin{aligned}\frac{\partial\mathbf{C}/\partial Y_M}{\partial\mathbf{C}/\partial Y_T} &= \left(\frac{\beta_M}{1-\beta_M}\right)^{\frac{1}{\xi}}\left(\frac{Y_M}{Y_T}\right)^{-\frac{1}{\xi}}(1+\mathcal{T}_c)^{\frac{1-\hat{\xi}}{\phi\hat{\xi}}}\left(1-\frac{1}{\phi}\frac{\partial\log(1+\mathcal{T}_c)}{\partial\log Y_M}\right)\\ &= \frac{q_M}{q_T}\left(1-\frac{1}{\phi}\frac{\partial\log(1+\mathcal{T}_c)}{\partial\log Y_M}\right)\end{aligned} \tag{21}$$

When the consumer fixed cost χ_c is positive, more modern service output reduces the implicit tax $\mathcal{T}_c$ on modern consumption, i.e. $\frac{\partial\log(1+\mathcal{T}_c)}{\partial\log Y_M}<0$. This is the key demand externality of the economy. Workers and firms do not internalize that by matching and producing a surplus over the worker's own-account work output, they contribute to aggregate demand by making the fixed cost of modern consumption less burdensome.

3.3 Modern Service Technology and the Big Push

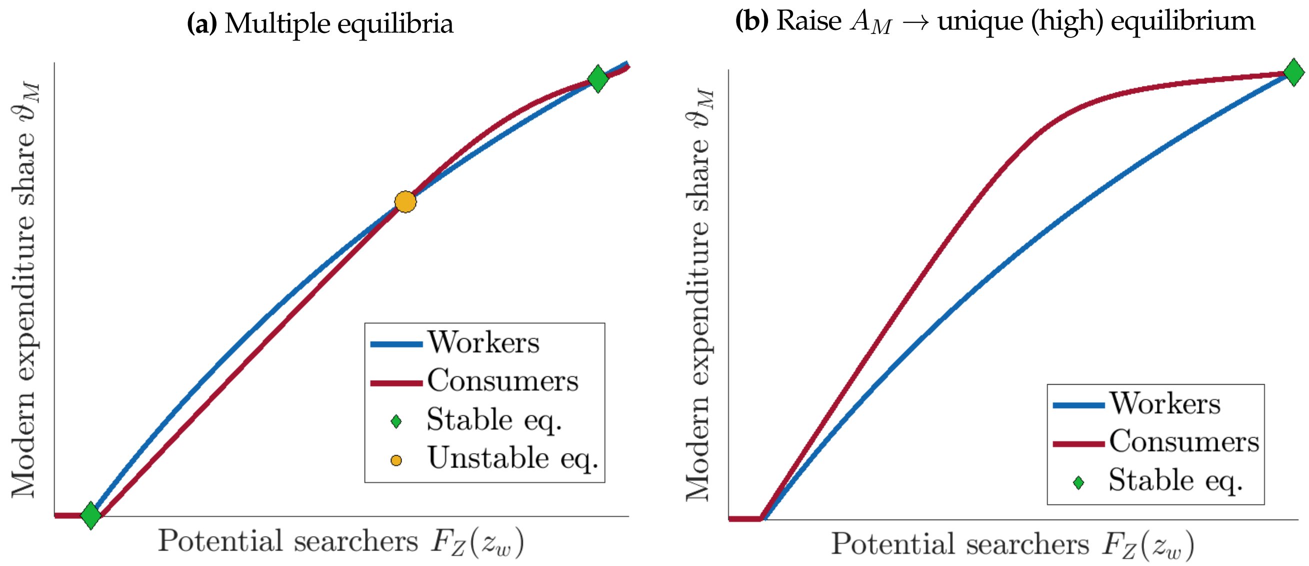
The characterization of equilibrium in terms of just two endogenous variables is a powerful tool to understand the economy's response to changes in fundamentals. In particular, I use it to study the effects of technological progress in modern service firms, i.e. an increase in A_M. Even small changes in this technology can generate large shifts in the economy's allocation.

An Extreme Case: Multiple Equilibria In extreme cases, increases in A_M can take the economy from multiple equilibria to a unique equilibrium. Figure 6 presents a parameterization of the model where workers and consumers respond very tightly to one another. In this case, multiple equilibria can arise with dramatically different modern service shares.

The economy in Figure 6a features two stable equilibria where the consumer choice function crosses the worker sorting function from above (indicated by green diamonds) and one unstable equilibrium (indicated by a yellow circle). I focus on the stable equilibria, since the economy almost surely converges to one of these in the long run. The stable equilibrium on the left is the *street market equilibrium* where the modern service sector does not operate: $\vartheta_M = H_M = 0$. Almost all workers remain in own-account work, except for a handful who work for wages producing goods. The stable equilibrium on the right is the *shopping mall equilibrium* where the service sector is almost entirely modern: ϑ_M is close to 1, and almost all workers are in wage employment.

The street market equilibrium is possible because if too few workers pursue wage employment,

Figure 6: From multiple to unique equilibrium



real purchasing power is too low to overcome the fixed cost of modern consumption, no consumers want to shop for modern services, and hence no modern service firms enter to hire workers, so that the returns to job search are low. If instead workers coordinated to work for wages, they could produce enough output to cover the fixed cost for consumers, inducing modern service firms to enter and creating demand for their own wage labor. Coordinating in this way would allow for the shopping mall equilibrium of the model.

Increasing A_M, as in Figure 6b, makes the shopping mall equilibrium unique without needing to coordinate. By increasing the supply of modern services even conditional on worker sorting z_w, higher A_M raises ϑ_M for any z_w, pushing up the consumer choice curve. This eliminates the lower equilibrium where the modern service sector does not operate and leads to a unique shopping mall equilibrium, where many firms enter and the economy is almost entirely modern.

The General Case: Amplification Even when the economy does not feature multiple equilibria, increases in A_M still generate amplified reallocation, because consumer and worker choice are highly complementary to one another. When workers make the choice to search for more productive wage work, they not only reduce the relative price of modern services, they also increase total income in the economy. Since demand for modern services is non-homothetic, rising income generates further demand shifts towards modern services, pulling even more workers and firms into the modern sector and leading to further income growth and reallocation. In this sense, the model is similar to the one in Buera et al. (2021): the same "big push" force that can generate multiple equilibria in extreme cases – income effects on demand, captured by $\chi_c > 0$ – generically acts as an amplification force.

To see this, consider the derivative of modern service labor H_M with respect to A_M. Because, by Proposition 1, H_M can be written as $H_M(z_w, \vartheta_M)$, its derivative satisfies:

$$\begin{aligned}\frac{dH_M}{dA_M} &= \frac{1}{1 - \frac{\partial \vartheta_M^*}{\partial z_w}\frac{\partial z_w^*}{\partial \vartheta_M}} \times \left(\frac{\partial H_M}{\partial z_w}\left(\frac{\partial z_w^*}{\partial A_M} + \frac{\partial z_w^*}{\partial \vartheta_M}\frac{\partial \vartheta_M^*}{\partial A_M}\right) + \frac{\partial H_M}{\partial \vartheta_M}\left(\frac{\partial \vartheta_M^*}{\partial A_M} + \frac{\partial \vartheta_M^*}{\partial z_w}\frac{\partial z_w^*}{\partial A_M}\right)\right) \\ &= \text{Amplification} \times \text{Direct effects of } A_M\end{aligned} \tag{22}$$

I provide a formal proof of this comparative static in Appendix B.4.

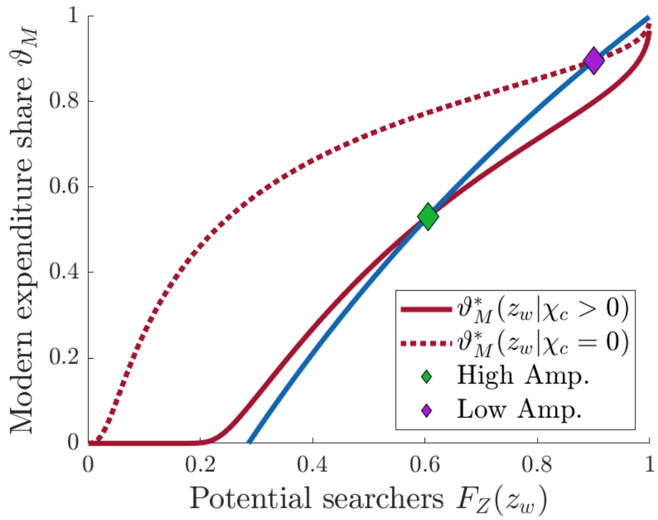
With reference to the graphical representation of the equilibrium, the "direct effects" of a change in A_M correspond to shifts of the Worker Sorting and Consumer Choice curves generated by A_M.[16] The "amplification" term corresponds to movement along the two curves to a new equilibrium point. It is larger the closer the two curves are to parallel, i.e. the closer $\frac{\partial \vartheta_M^*}{\partial z_w}\frac{\partial z_w^*}{\partial \vartheta_M}$ is to 1.

The income effect from the consumer fixed cost $\chi_c > 0$ amplifies all comparative statics of the economy because it increases $\frac{\partial \vartheta_M^*}{\partial z_w}$, the impact of the worker search margin on demand for modern services. It is possible to show this analytically, but more straightforward to see it graphically. Figure 7 presents optimal consumer choice $\vartheta_M^*(z_w)$ under two parameterizations. The solid red line represents the generic case of non-homothetic demand, where the fixed cost χ_c is positive. The dotted red line represents the homothetic benchmark where $\chi_c = 0$.

The fixed cost has two effects. First, it drags down the consumer choice curve and generates a positive-measure region of $F_Z(z_w)$ where no consumers shop in the modern sector. In this region, there is not enough output generated to cover the fixed cost of modern shopping. Second, χ_c increases the slope of $\vartheta_M^*(z_w)$ outside the region where $\vartheta_M^*(z_w) = 0$. Why? Because the non-homotheticity adds to the response of consumers to worker job search. As more workers search and increase the value of output, they add to the demand for modern services. This effect is absent when demand is homothetic. As a result of the steeper slope of $\vartheta_M^*(z_w)$, amplification is greater in the equilibrium when $\chi_c > 0$, marked by a green diamond, than when $\chi_c = 0$, marked by a purple diamond.

[16] It turns out that in fact the Worker Sorting curve does not shift in response to A_M, $\frac{\partial z_w^*}{\partial A_M} = 0$. This is because, holding ϑ_M fixed, A_M does not alter the skill prices facing workers, due to the Cobb-Douglas nature of final production.

Figure 7: Optimal consumer choice with and without non-homotheticity χ_c



3.4 Modern Service Technology and Service-Led Growth

How does the amplified reallocation generated by increases in A_M lead to aggregate gains in the economy? Using the definition of consumption in (19) and aggregate production functions for intermediates, it is possible to break down the sources of consumption growth generated by increases in A_M. In particular, $\frac{d\log\mathbf{C}}{d\log A_M}$ can be decomposed into a mechanical gain and efficiency gains:

$$\frac{d\log\mathbf{C}}{d\log A_M} = \underbrace{\phi\vartheta_M}_{\text{Mechanical Gain}} - \underbrace{\vartheta_M \frac{d\log(1+\mathcal{T}_c)}{d\log A_M}}_{\text{Efficiency Gain: Consumption}} + \underbrace{w_f \frac{dH_w}{d\log A_M} + w_T \frac{dH_T}{d\log A_M}}_{\text{Efficiency Gain: Reallocation}} \tag{23}$$

Mechanical Gain Increasing A_M generates mechanical growth proportional to the share of expenditure $\phi\vartheta_M$ allocated to modern services. In a completely frictionless economy with no search frictions or consumer fixed costs, this would be the only effect of A_M. But in this case, it also generates efficiency gains: from reducing the implicit tax on modern consumption, and from reallocating labor to more productive uses.

Efficiency Gain: Consumption When consumers face fixed costs $\chi_c > 0$, the growth of A_M reduces the endogenous wedge $\mathcal{T}_c$ on modern consumption. By reducing the bite of the fixed cost χ_c, growth increases the efficiency of modern expenditure – more of it translates into consumption. This gain is proportional to the modern expenditure share ϑ_M.

Efficiency Gain: Reallocation Most importantly, raising A_M pulls workers out of the traditional sector and into wage work as the modern sector becomes more productive. That is, it increases H_w and decreases H_T. Because of frictions κ_w, this reallocation generates a strictly positive gain. In particular:

$$w_f \frac{dH_w}{d\log A_M} + w_T \frac{dH_T}{d\log A_M} \geq w_f \kappa_w \frac{dH_w}{d\log A_M} > 0, \tag{24}$$

so that the efficiency gain from reallocation is bounded away from zero when $\kappa_w > 0$. This result follows from the fact that workers on the margin z_w are more productive in wage work: $w_f > w_T z_w$. The Lewis-style gain it implies is a powerful driver of service-led growth. It becomes even more powerful when one recalls that $\frac{dH_M}{d\log A_M}$ is amplified by the "big push" forces of the previous subsection. In particular, one can use Equation (22) to write reallocation gains as a product of direct "Lewis gains" and amplification, i.e. "big push" gains:[17]

$$\begin{aligned} \text{Reallocation gains} &\geq \frac{1}{1 - \frac{\partial \vartheta_M^*}{\partial z_w} \frac{\partial z_w^*}{\partial \vartheta_M}} \times w_f \kappa_w \left(\frac{\partial H_w}{\partial z_w} \frac{\partial z_w^*}{\partial \vartheta_M} + \frac{\partial H_w}{\partial \vartheta_M} \right) \frac{\partial \vartheta_M^*}{\partial \log A_M} \\ &= \text{Big Push} \times \text{Direct Lewis Gains} \end{aligned} \tag{25}$$

When the labor market is subject to frictions $\kappa_w > 0$, and demand is subject to income effects $\chi_c > 0$, growth of modern service technology has the potential to unleash significant service-led growth. Rather than being mechanical, a large share of service-led growth comes from reallocation gains in the style of Lewis (1954), amplified by Big Push forces in the style of Buera et al. (2021).

4 Estimation and Validation

The previous section explores the theoretical effects of technological progress in modern services. This section estimates its real-world effects on the Brazilian economy. I pair data on the Brazilian service transformation with the fully parameterized model to estimate the growth of modern service technology for every microregion in Brazil from 2000-2010. The rise of modern service employment is not sufficient to identify this growth. As the theory shows, modern services could have expanded due to an abatement of labor market frictions κ_w, or due to general income growth from outside the service sector. To disentangle these explanations, I leverage additional data and map the theory onto each individual microregion.

[17] Here I have also used the fact that $\frac{\partial z_w^*}{\partial \log A_M} = 0$.

More formally, I treat each microregion in each Census year as an independent instance of the single economy described in the theory. This is consistent with 2000 and 2010 as separate steady states of the model, and with each microregion as a closed economy – or, as I show in Appendix C.2, a small open economy with free trade in intermediate goods.[18] Each microregion-year rt has a set of structural parameters in common, but is characterized by three fundamentals that I allow to vary over space and time: worker search costs $\kappa_{w,rt}$, intermediate goods technology $\Psi_{X,rt}$, and modern service technology $A_{M,rt}$. These three fundamentals can be estimated separately for each microregion-year, using the data and methods I outline below. Growth of modern service technology corresponds to an increase of microregion-level A_M from 2000 to 2010.

4.1 Data

I use three main data sources: the Brazilian Census, to obtain local measures of modern service employment, modern and traditional service earnings, and aggregate income; consumer expenditure data, to measure the strength of non-homotheticity; and worker panel data, to measure employment and earnings dynamics. To process all data in the paper, I use the Data Zoom package developed by the Economics Department at PUC-Rio.

Brazilian Census I use the long form of the Census for the years 2000 and 2010. For each individual, the long-form Census includes information on employment status, industry of employment, and take-home income, as well as geographic information on the municipality and microregion where the individual lives.

Using person weights provided by IBGE, I aggregate individual information up to the microregion level to extract three statistics for each microregion-year: the share of service workers in modern employment, the cross-sectional wage premium of modern service work, and aggregate income per capita.[19] These three statistics are crucial to separately identify modern service technology from labor market frictions and productivity growth outside services. I also use the Census for aggregate national statistics that I use in the calibration, such as the average employment share of services and the average within-microregion dispersion of modern and traditional earnings.

[18] In this case, the "productivity" of the goods sector depends in part on the microregion's terms of trade.

[19] As in Section 2, "services" refers specifically to consumer services: retail and wholesale trade, hospitality and food service, transportation, and other personal services. Modern employment refers to wage employment with pension contributions.

Consumer expenditure survey (POF) I use the 2008 vintage of Brazil's consumer expenditure survey to estimate the modern shopping fixed cost χ_c. The POF records the type of physical establishment where each household purchase took place, as the "location of purchase."[20] I group these locations into "modern" and "traditional" final consumption establishments using the classification developed by Bachas et al. (2023), and aggregate modern and traditional expenditure over the past year for each household.[21] A household's modern share of expenditure is total modern expenditure divided by the sum of modern and traditional expenditure, as seen in the histogram in Figure 4b. Besides categorized expenditure, the POF also collects information on household income, demographics, and geography.

Worker panel survey (PME) I use the PME for the years 2002-2016 as a source of panel data on workers. It is a rotating monthly panel survey of urban workers in six large Brazilian metropolitan areas that records employment status and income.[22] I use the PME to estimate the worker-level covariance between modern and traditional earnings, which I use to calibrate the correlation between own-account and wage employment ability. With the PME I can also compute transition rates between modern and traditional employment, which I use to calibrate the matching function efficiency and the exogenous separation rate δ in the labor market.

4.2 Functional Forms and Extensions

Matching function I model the labor market matching function as Cobb-Douglas, so that $\lambda(\theta) = \overline{\lambda}\theta^{1-\alpha}$.

Heterogeneity in wage work ability I now add a second dimension of worker heterogeneity, to better account for differences in wage earnings across workers. In the quantitative model, workers are heterogeneous in an absolute ability scalar h that multiplies both wage work and own-account ability. Workers of type (z, h) supply h efficiency units of labor to wage work and zh to own-account work, so that z still captures comparative advantage in own-account work. For tractability, I assume

[20] Location of purchase is not recorded for certain recurring payments such as rent, utilities, and loan repayments.

[21] Following Bachas et al. (2023), I classify specialized shops, large stores, service institutions, and entertainment institutions as sources of modern consumption. Traditional consumption comes from non-market sources, vendors without a storefront, convenience/corner shops, individual service providers, and informal entertainers.

[22] Workers are interviewed up to eight times over a period of sixteen months. In the absence of attrition, workers are interviewed for four consecutive months, then uninterviewed for eight consecutive months, then interviewed again for four more consecutive months.

that h also multiplies vacancy and search costs for type (z,h) workers to be $\kappa_v h$ and $\kappa_w h$. This implies that all relevant search decisions are still functions of only z – in particular, market tightness $\theta(z)$ depends only on z, the sorting threshold z_w is independent of h, and employee wages are $w_e(z)h$. I parameterize the distribution of (z,h) across workers as joint log-normal:

$$\begin{pmatrix} \log z \\ \log h \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_z \\ \mu_h \end{pmatrix}, \begin{pmatrix} \sigma_z^2 & \varrho\sigma_z\sigma_h \\ \varrho\sigma_z\sigma_h & \sigma_h^2 \end{pmatrix} \right) \tag{26}$$

Since μ_z and μ_h are isomorphic to aggregate technology levels, I set them so that zh and h each have mean 1 in the population.

4.3 Estimation Strategy

I index each of the 557 microregions in Brazil by $r=1,...,557$ and consider them in two years, $t=2000$ and $t=2010$, treating each microregion-year as a separate economy. The model is parameterized by a set of structural parameters that I assume to be constant over time and space. Consumer preferences depend on four parameters: utility weight on modern services β_M, modern/traditional elasticity of substitution ξ, service Cobb-Douglas weight ϕ, and discount factor ρ. The worker ability distribution is characterized by dispersion of absolute ability σ_h, dispersion of relative own-account ability σ_z, and correlation between the two ϱ. Search and matching parameters are the matching function efficiency $\overline{\lambda}$, matching elasticity α, match separation rate δ, and worker bargaining weight η. Finally, I take two technological parameters to be time- and space-invariant: traditional technology ψ_T and consumer fixed cost χ_c.

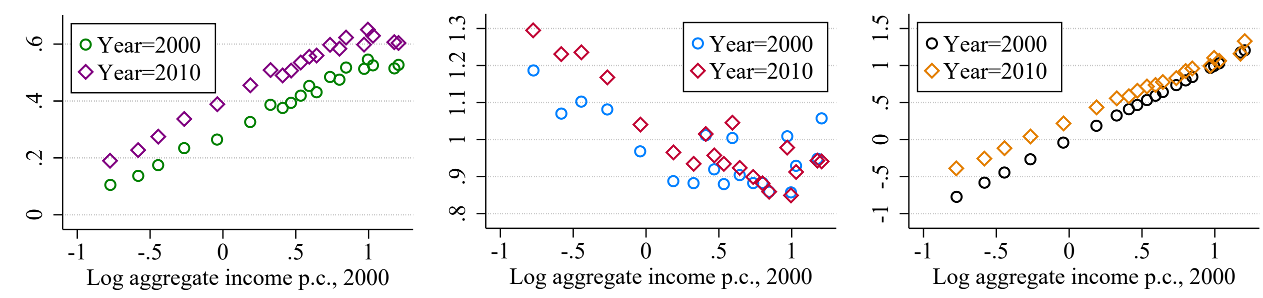
Each microregion-year is characterized by three local fundamentals: worker search cost $\kappa_{w,rt}$, goods technology $\Psi_{X,rt}$, and modern service technology $A_{M,rt}$. The goods technology can be thought of as a neutral source of income growth that affects the economy's allocation only through income effects on demand.

My estimation strategy combines structural accounting with moment matching. Specifically, I estimate local primitives $\kappa_{w,rt},\Psi_{X,rt},A_{M,rt}$ to match the three microregion-level moments I take from the Census: the share of service workers employed in the modern sector, denoted $\hat{L}_{M,rt}$, the ratio of average modern earnings to average traditional earnings within services, denoted $\hat{w}_{M,rt}$ (the modern wage premium), and aggregate income per capita, denoted $\overline{y}_{rt}$.

In Figure 8, I present binned scatters showing how the three local target moments vary across microregions of different income levels and over time. Panel 8a is a binned scatter version of Figure 1b, showing service modernization over time and space. Panel 8b presents the cross-sectional modern wage premium in 2000 and 2010 against the log of per capita income in 2000. Richer microregions tend to have a lower wage premium, although the relationship is noisier than for employment shares. And for a large share of microregions, the modern wage premium increased from 2000 to 2010. Panel 8c plots aggregate income in 2000 and 2010 against income in 2000. The 2000 points therefore follow the 45-degree line, while the 2010 points show that Brazil experienced regional convergence: the poorest microregions had the highest 2000-2010 income growth.

Figure 8: Identifying Variation: Employment Shares, Wage Premium, and Income in 2000 and 2010

(a) Modern share of services $\hat{L}_{M,rt}$ **(b)** Modern wage premium $\hat{w}_{M,rt}$ **(c)** Log aggregate income p.c. $\overline{y}_{rt}$



Note: Data from the 2000 and 2010 Brazilian Census. Aggregate income per capita is denominated in units of the real 2000 Brazilian minimum wage. All aggregation within microregions uses individual person weights as provided by the Census, and microregions are weighted by population in the binned scatters.

Conditional on structural parameters, there is a direct mapping between these three moments and local fundamentals. I use a sequential procedure to estimate the fundamentals. First, I use the modern wage premium $\hat{w}_{M,rt}$ and modern share of service employment $\hat{L}_{M,rt}$ to measure $\kappa_{w,rt}$ and the modern expenditure share $\vartheta_{M,rt}$. One way to think of the wage premium is as a measure of the surplus that wage work provides over own-account work for service workers. As seen in the theory section, this surplus could be high either because consumers demand a lot of modern services (high $\vartheta_{M,rt}$), or because the worker search costs $\kappa_{w,rt}$ are high, so that wage work must pay a large premium to compensate. If the wage work surplus is high because of demand then the modern share of service employment will be high; if it is high because of worker search costs, then the modern share of services will be low. $(\hat{L}_{M,rt},\hat{w}_{M,rt})$ therefore jointly identify $(\vartheta_{M,rt},\kappa_{w,rt})$. Of course, when using the observed wage premium it is necessary to adjust for selection, which I do with the model of sorting into wage work coupled with estimates of the worker ability distribution.

Next, having estimated $(\vartheta_{M,rt}, \kappa_{w,rt})$, the worker sorting threshold $z_{w,rt}$ and allocation of efficiency units $(H_{M,rt}, H_{T,rt}, H_{X,rt})$ follow mechanically from the model. Given the allocation of efficiency units, there is a one-to-one mapping between goods technology $\Psi_{X,rt}$ and local aggregate income $\overline{y}_{rt}$. I therefore use $\overline{y}_{rt}$ to estimate $\Psi_{X,rt}$.

In the final step, I infer the value of $A_{M,rt}$ that can explain the modern service share $\vartheta_{M,rt}$. All other determinants of consumer choice – the efficiency unit allocation and technology in other sectors – are known by this last step. $A_{M,rt}$ is therefore the fundamental that the estimation uses to explain the part of the modern service share that cannot be rationalized by other economic forces.

4.4 Location-Invariant Parameters

The estimation of regional fundamentals outlined in 4.3 holds for any given vector of location-invariant parameters. To estimate these parameters, I first pre-set some of them as a normalization and assign values taken from the literature to some others. For the remainder, I either use direct estimates from the data or calibrate them using the method of simulated moments. I lay out all parameter values and their target moments and/or sources in Table 1. The model is exactly identified so that by design it matches all target moments.

Table 1: Calibrated Parameters

Parameter	Description	Value	Target Moment	Moment Value
χ_c	Modern consumption fixed cost	0.237	Engel curve slope (IV)	0.14
ϕ	Cobb-Douglas weight of services	0.60	Service share of emp.	0.60
σ_h	Dispersion of $\log h$	0.39	SD($\log y \vert M$)	0.55
σ_z	Dispersion of $\log z$	1.05	SD($\log y \vert T$)	0.89
ϱ	Correlation of $\log h$,$\log z$	−0.24	Corr($\log y \vert M$,$\log y \vert T - \log y \vert M$)	−0.172
$\overline{\lambda}$	Matching function efficiency	16.9	Trad.-modern transition rate	0.092
δ	Modern match separation rate	0.059	Modern-trad. transition rate	0.059
ρ	Discount rate (monthly)	0.007	Brazil real interest rate (2010)	8.9%
$\kappa_{w,rt}$	Worker search friction	rt-specific	Modern service wage premium	
$A_{M,rt}$	Technology in M	rt-specific	Modern share of service emp.	
$\Psi_{X,rt}$	Technology in X	rt-specific	Aggregate income per capita	
Parameter	**Description**	**Value**	**Source**	
ψ_T	TFP in T	1.0	Normalization	
β_M	Preference weight for M	0.5	Normalization	
κ_v	Vacancy posting cost	1.0	Normalization	
ξ	M-T elasticity of substitution	3.97	Feng et al. (2024)	
α	Matching function elasticity	0.5	Petrongolo and Pissarides (2001)	
η	Worker bargaining weight	0.5	Petrongolo and Pissarides (2001)	

Pre-set parameters I assign pre-set values to six parameters. ψ_T and β_M are not separately identified from other parameters, so I normalize them to 1 and 0.5 respectively. Similarly, the vacancy posting cost for employers in the search and matching model is not separately identified from the matching function efficiency $\overline{\lambda}$, so I normalize it to 1. I then pre-set three additional parameters based on estimates from the literature. For the elasticity of substitution between modern and traditional consumption, I use the estimate of 3.97 found by Feng et al. (2024). I set the matching function elasticity and worker bargaining weight to both equal 0.5, following Petrongolo and Pissarides (2001), which allows me to solve for $\theta(z)$ analytically.

Direct estimates I estimate two parameters, δ, the wage employment separation rate, and ρ, the discount rate, directly from the data. δ corresponds exactly to the average monthly transition rate of modern wage workers into traditional own-account work. I estimate this transition rate at a value of 5.9% on average in the PME. For ρ, I use the 2010 real interest rate on deposits in Brazil. According to the IMF, this rate was 8.9% in annual terms. Since the model treats a month as one unit of time, I set $\rho = 0.007$ to match the corresponding monthly rate of 0.7%.

Labor market calibration I use the method of simulated moments and jointly estimate five parameters to match five moments from the aggregate Brazilian labor market. The five parameters are (i) σ_h, the dispersion of log absolute ability h, (ii) σ_z, the dispersion of log relative ability in traditional services z, (iii) ϱ, the correlation between $\log h$ and $\log z$, (iv) $\overline{\lambda}$, the efficiency of the matching function in the market for modern work, and (v) ϕ, the Cobb-Douglas weight of services in the composite consumption good. I target five moments from the aggregate Brazilian labor market: the dispersion of log earnings among modern and traditional service workers respectively, the monthly traditional-to-modern transition rate, the correlation between log modern earnings and the individual modern earning premium – i.e. how much more an individual "switcher" worker makes in the modern sector – and the service share of total employment.

For the MSM procedure, I simulate the same moments in the model for a "representative microregion" that has the modern employment share, modern wage premium, and income per capita of Brazil's aggregate economy in 2010. For a given guess of the structural parameters, I estimate κ_w, Ψ_X, A_M for the representative microregion, then simulate the relevant moments. I iterate on guesses of the parameters until the simulated moments exactly match the target moments.

While the parameters must be estimated together, each clearly maps to a particular observable moment. σ_h governs the dispersion of log earnings among modern workers, and σ_z governs the same dispersion among traditional workers; I estimate both of these moments in the Census after stripping out microregion fixed effects from earnings. $\bar{\lambda}$ primarily determines the rate at which workers transition from traditional to modern work, which I estimate in the PME. The correlation coefficient ϱ requires data from the PME on workers who transition between modern and traditional work; I calibrate it to match the across-worker correlation between log modern earnings, which mostly reflects $\log h$, and the difference between log traditional earnings and log modern earnings, which mostly reflects $\log z$. ϕ primarily determines the overall share of employment in services as opposed to tradable goods as taken from the 2010 Census.

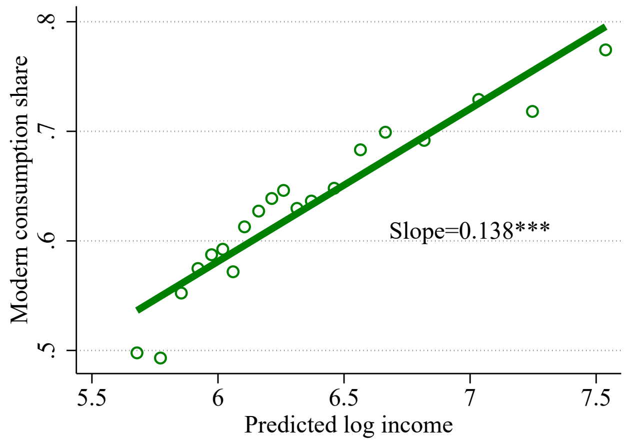
Consumer fixed costs χ_c The last parameter I calibrate is χ_c, the size of the fixed cost facing consumers, which governs the strength of income effects. I therefore calibrate it to match the slope of the Engel curve for modern service consumption. To adjust for potential attenuation bias of the Engel estimates due to measurement error, I instrument for log income using a set of 3-digit occupation fixed effects and use the corresponding IV estimate for the slope of the Engel curve, which is a statistically significant 0.138. I use location fixed effects to strip away differences across individuals that are potentially due to geography.[23] I present the instrumented Engel curve below in Figure 9. The estimated value of $\chi_c = 0.237$ is substantial: for Brazil in the aggregate in 2010, the impact of this fixed cost on consumer choice is equivalent to that of a 16% proportional tax on modern consumption.

4.5 Results: Local Productivity and Search Costs

In Figure 10, I plot binned scatters of two key estimated fundamentals – modern service productivity A_M and worker search costs κ_w – against the log of microregion-level income per capita in 2000. Panel 10a plots the log of A_M on the same scale as log aggregate income. The covariance of $\log A_M$ with aggregate income is remarkably weak: the gap in modern productivity between the richest and poorest regions of Brazil is a small fraction of the gap in income. Despite the enormous differences in modern service employment between rich cities like Sao Paulo and impoverished rural areas, modern firms are not massively more productive in rich regions. On the other hand, $\log A_M$ does have a clear time-series pattern, increasing by about 0.24 log points from 2000 to 2010

[23] In this case, location fixed effects are for primary sampling units (PSUs).

Figure 9: Instrumented Engel Curve

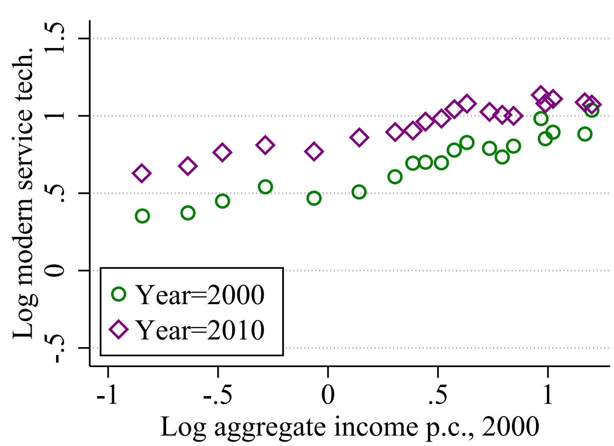


Note: Data from POF; expenditure classification follows Bachas et al. (2023). I control for the same household characteristics as in the "OLS" Engel curve of Figure 4a but instrument for income using 3-digit occupation fixed effects, to eliminate attenuation bias from measurement error.

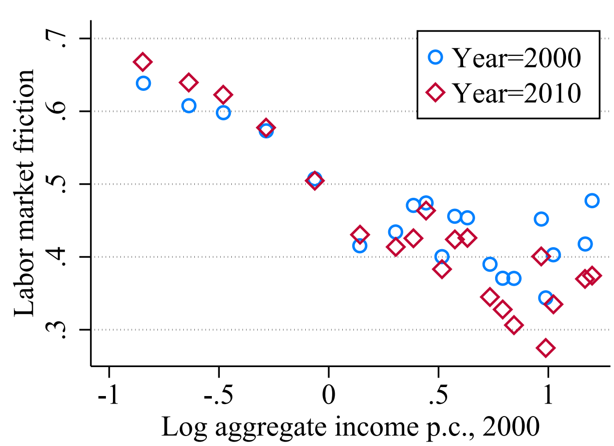
in a balanced way across Brazil. These results suggest that there is a strong pan-Brazil component to the modern service technology A_M that may stem from economy-wide changes like advances in communications. In Appendix C.1, I show that these patterns of A_M stand in sharp contrast to the goods technology Ψ_X, which varies more substantially in the cross-section and displays significant convergence from 2000-2010.

Figure 10: Estimated Primitives by Microregion Income

(a) Modern service technology A_M



(b) Search costs κ_w



Note: Estimation results for $\log A_M$ and κ_w, using the method described in 4.3.

Panel 10b shows the estimated worker search cost κ_w across microregions in 2000 and 2010. Frictions do follow a significant gradient with aggregate income: they are substantially more severe in poor microregions, which helps explain why poor parts of Brazil have low modern employment despite a high modern wage premium. Unlike A_M, search frictions show no signs of substantial improvement from 2000 to 2010 – in some parts of Brazil, they actually got more severe. These estimates follow naturally from the fact that modern service employment expanded from 2000 to 2010 while the modern wage premium rose rather than fell.

4.6 Untargeted Moments

By design, the estimation strategy exactly matches the target moments. Now I test it against untargeted moments. The model hinges on non-homothetic consumer demand and worker sorting between permanent own-account work and search. I document that the model performs very well at capturing moments relevant to these key consumer and worker decisions.

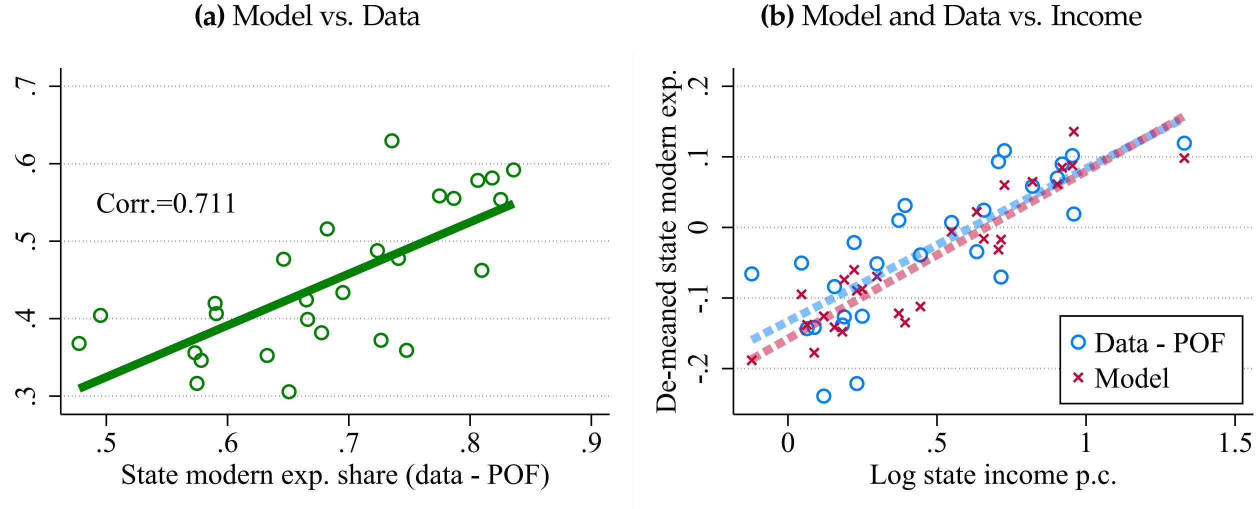
Modern expenditure shares The model uses labor market data to infer the modern expenditure share $\vartheta_{M,rt}$ in every microregion in each year. The consumer expenditure data does not report microregions, but I can use it to compute modern expenditure shares at the state level, then aggregate model-generated data to the state level to compare.

In Figure 11, I compare model and data estimates of the modern expenditure share. Panel 11a directly plots model estimates against data estimates. There is a level difference where the model systematically underestimates the modern expenditure share as measured in the data. This would happen if some establishments that are classified as modern employ "traditional" workers. For example, a bakery run by a self-employed entrepreneur might show up as modern expenditure because it is a specialized store, but in labor market data I would classify the baker as a traditional service worker because she is self-employed. Despite the level difference, variation in the model aligns quite well with variation in the data, and the two measures have a strong positive correlation of 0.71.

In Panel 11b, I compare de-meaned state-level expenditure shares in the model and data against state-level income. The model does very well at capturing the systematic co-variation of modern expenditure and income: the slopes are almost identical between the data and the model.

Own-account work and income In my model, individuals could be own-account workers for two reasons. Either they are gifted entrepreneurs, or search frictions push them into own-account work for subsistence income. These two types of own-account workers are visible in the data on

Figure 11: State-Level Modern Expenditure Share $\vartheta_{M,s}$: Model vs. Data (POF)



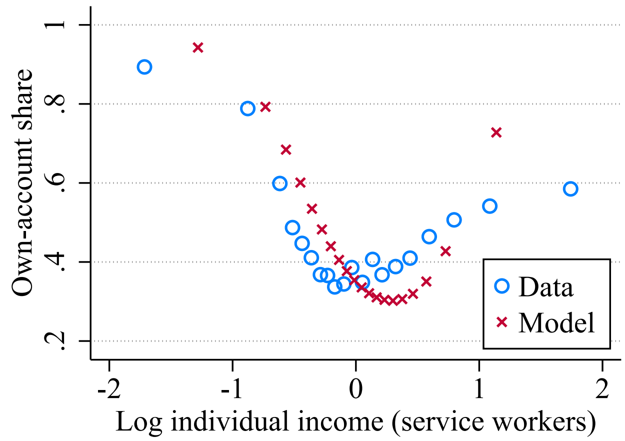
Note: ϑ_M in the model is estimated at the microregion level and then aggregated up to the state level. In the data, I use the 2008 POF with household weights to compute the modern share of all household expenditures within a state.

individual earnings and own-account employment status. In particular, in Figure 12, I examine own-account employment over the individual income distribution of service workers. I place these workers into bins based on their income, after residualizing on microregion fixed effects. Then within each income bin, I plot the probability that a worker in that bin is employed in own-account work, also after residualizing on microregion fixed effects.

Consistent with the model, Figure 12 shows that the relationship between own-account employment and income – plotted by blue dots – has a pronounced U-shape. Among the lowest-earning service workers, over 80% are in own-account work – these are subsistence workers. In the middle of the income distribution this number drops below 40%. But in the right tail among the richest workers, almost 60% are own-account workers – the gifted entrepreneurs.

In red X-marks, I show the same relationship within my model, using the model-generated income and employment distribution for a representative microregion that has the same modern employment share and wage premium as the aggregate Brazilian economy. The model distribution has the same pronounced U-shape as is present in the data. It puts more mass on the left side of the U, suggesting that a higher share of own-account workers are subsistence workers than are gifted entrepreneurs.

Figure 12: Own-account employment vs. income among service workers



Note: Both model and data distribution have microregion fixed effects stripped out. Specifically, in the data I regress both log individual income and own-account employment on fixed effects for each microregion, then I plot the individual-level residuals against one another. Data source is the Census.

Natural experiment: trade liberalization Finally, I offer additional validation from a natural experiment. The central driver of amplification in the model is that rising incomes shift aggregate demand to modern services even when the relative productivity of modern services is held constant. An ideal experiment to test this feature would be a shock to Ψ_X: this shock holds modern productivity fixed but should still affect employment and expenditure shares due to income effects.

The Brazilian context provides a natural experiment of just this type. I use Brazil's unilateral trade liberalization in the early 1990s, when the government cut tariffs on a large set of tradable goods. Microregions were differentially exposed to import competition based on their industry mix, generating quasi-random variation. A large literature has studied the local impacts of this exposure, including Kovak (2013), Dix-Carneiro and Kovak (2017), Dix-Carneiro and Kovak (2019), Ponczek and Ulyssea (2022), Dix-Carneiro et al. (2021), and Felix (2021).

Trade liberalization generated exogenous microregion-level variation in the price of locally produced tradable goods. In Appendix C.2 I show that this shock is isomorphic to a shock to $\Psi_{X,rt}$. In the model, when $\chi_c > 0$ so that demand is non-homothetic, a shock to $\Psi_{X,rt}$ acts as a demand shock for modern services. Exogenous increases in $\Psi_{X,rt}$ should therefore lead to a higher modern share of service employment. They should also increase the modern wage premium within services, to draw more workers into wage employment in the modern sector.

To test this prediction, I measure local exposure to import competition using the "regional tariff reductions" computed by Dix-Carneiro and Kovak (2017). I use tariff reductions as an instrument for an exogenous income shock originating outside the service sector. Using two-stage least-squares, I estimate the impact of this shock on microregion-level modern employment and the modern wage premium among service workers in 2000, the first Census year following the shock. In all empirical regressions, I control for local aggregate income, modern service share, and modern wage premium in the Census year prior to the shock (1991).

I present results from these regressions in Table 2. The results confirm one of the model's central predictions: even though the income shock came from outside the service sector, it acted as a demand shock on modern services. A 10% trade-induced increase in income induces 0.5% of service workers to switch to modern services, and increases the modern wage premium in services by 1.9%.

Table 2: Impact of trade-driven income shock on local service sector in 2000

	Modern share of service emp.	Modern wage premium in services
Log income p.c.	0.053^{***}	0.186^{**}
	[0.014,0.092]	[0.000,0.373]
Observations	389	389
State FE	Yes	Yes
Lag Controls	Yes	Yes
Log Income IV	Tariff shock	Tariff shock
1st Stage F-Stat	80.9	80.9

Note: 95% CI from robust standard errors in brackets. Outcomes are in 2000, the first Census year post-trade liberalization. All regressions control for 1991 modern employment share, log income, and modern wage premium. Microregions are weighted by 1991 population. I present the first stage regression in Appendix C.2.

The results in Table 2 are not only qualitatively in line with my theory; my model can generate quantitatively similar effects. To show this, I estimate local model primitives $\kappa_{w,rt}$,$A_{M,rt}$,$\Psi_{X,rt}$ for all relevant microregions in 1991 using the Census data from that year.[24] For each microregion, I take the trade-induced income shock predicted by the first stage of the two-stage least squares regression above. I reproduce this shock in the model by finding the counterfactual change in $\Psi_{X,rt}$ that would generate the same shock to income. Then I re-simulate the model under this $\Psi_{X,rt}$ and compute the new allocation and wage premium within services. Finally, I reproduce the regressions of Table 2 using the model-simulated data for each microregion.

In the model, a 10% trade-induced income shock pulls 0.24% of service workers into the mod-

[24]Note that there were fewer microregions in 1991; based on the Dix-Carneiro and Kovak (2017), for this exercise I use microregions with boundaries that are consistent back to 1970.

ern sector and raises the modern wage premium by 0.41% on average across microregions. The model is slightly conservative relative to the empirical results, but well within the 95% confidence intervals of the empirical estimates. It leaves open the possibility for other mechanisms, such as production linkages, to partially explain the impact of trade on the service sector (Dix-Carneiro et al., 2021). Still, the consumer demand channel laid out by the model explains about half of the quantitative change in service composition generated by trade shocks. I conclude that the model is qualitatively valid and, if anything, quantitatively understates the demand-driven impact of aggregate income growth on service transformation.

5 Service-Led Growth: Catalyst, Amplification, and Reallocation

Finally, with the estimated and validated model in hand, I use it to quantify the drivers of service-led growth and transformation within Brazil from 2000 to 2010. I show that modern service technology was the main catalyst that set off the transformation, and that amplification forces from non-homothetic demand and reallocation gains due to frictions are both quantitatively significant.

5.1 Modern Service Technology as Catalyst

Growth of the modern service technology A_M was the key fundamental that triggered service transformation and service-led growth in Brazil. To show this, I start from the baseline estimates of local fundamentals in 2000, $\kappa_{w,r2000}, \Psi_{X,r2000}, A_{M,r2000}$. Then I change only A_M from 2000 to 2010, re-simulate the model, and compare it against the full model where all three fundamentals are free to change. I do this for two outcomes. First, service transformation: the observable shift of service workers into modern employment. Second, service-led growth, the model-implied change in the log of real service consumption per service worker:

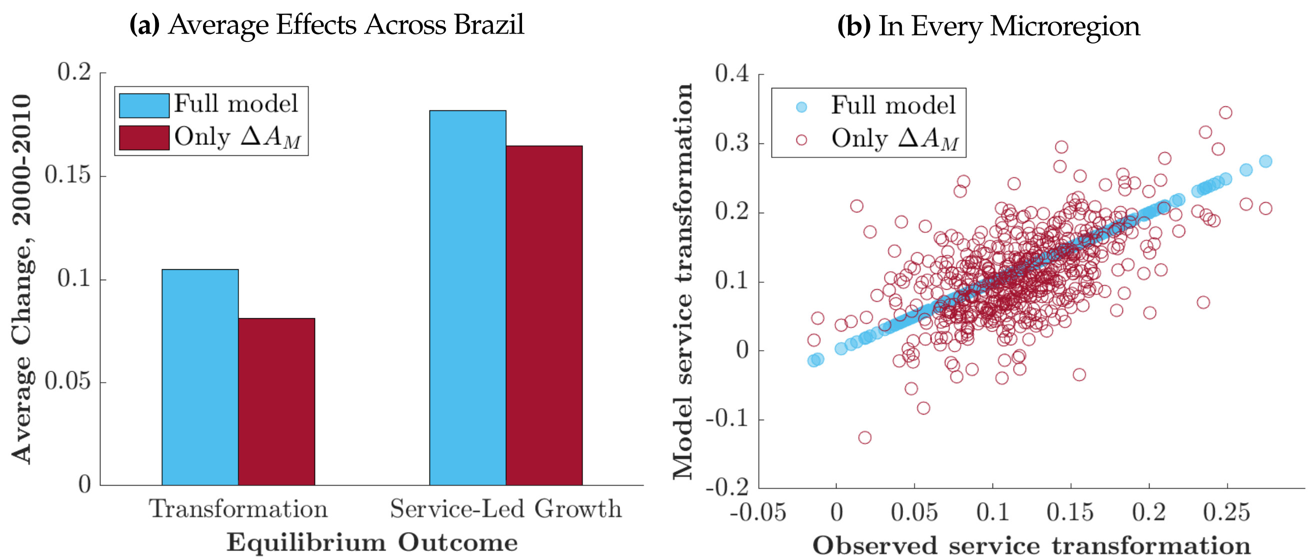
$$\begin{aligned} \text{Service Transformation} &= \Delta\left(\frac{L_M}{L_M+L_T}\right)_{2000,2010} \\ \text{Service-Led Growth} &= \Delta\log\left(\frac{\mathbf{C}_S}{L_M+L_T}\right)_{2000,2010}, \end{aligned} \tag{27}$$

where L_M is employment in modern services, L_T is employment in traditional services, and $\mathbf{C}_S$ is the CES composite of traditional and modern services.

I present the results in Figure 13. Panel 13a plots population-weighted average changes across all Brazilian microregions, for both the full model and the counterfactual with only A_M growth. Modern service technology clearly accounts for the vast majority of changes in the service sector from 2000 to 2010. Average service transformation over this period was 10.5 percentage points; the counterfactual with only A_M growth accounts for 8.1 percentage points of transformation. Meanwhile, service-led growth was 18.2% in the full model and 16.5% in the A_M-only counterfactual.

Panel 13b presents a scatter plot of the observed and model-implied service transformation for every microregion in Brazil. The blue dots correspond to the service transformation implied by the full model – by construction, they follow the 45 degree line, exactly matching actual service transformation. The red dots are the transformation in each microregion if only A_M had changed. This counterfactual is, of course, noisier, but is clearly correlated with the actual transformation.

Figure 13: Effects of Growing Modern Service Technology



The results from Figure 13 are clear: modern service technology alone can explain the vast majority of service transformation and growth within Brazil. As technology available to supermarkets and restaurants grew, more of these modern service firms entered. This process induced large productivity gains as traditional micro-entrepreneurs closed down autonomous production and took up new, more productive jobs in modern firms.

5.2 Non-Homothetic Demand Implies Amplification

Next, I assess the quantitative importance of the demand externality imposed by non-homothetic demand when the fixed cost $\chi_c > 0$. I do this by simulating the development of the Brazilian service sector in the counterfactual world of homothetic demand, where $\chi_c = 0$. Specifically, I assume χ_c to be zero and re-estimate counterfactual local fundamentals $\kappa^{cf}_{w,r2000}$,$A^{cf}_{M,r2000}$,$\Psi^{cf}_{X,r2000}$ for every microregion in 2000 under that assumption. Then I feed the actual measured change in primitives $\Delta\kappa_{w,r}$,$\Delta A_{M,r}$,$\Delta\Psi_{X,r}$ into the homothetic model and simulate the counterfactual change in the model. Again I do this for the two key service-sector outcomes: service transformation and service-led growth.

Non-homothetic demand $\chi_c > 0$ generates quantitatively important amplification of service transformation. The overall modern share of service employment grew by a population-weighted average of 10.5 percentage points on average over microregions. If demand was homothetic, service transformation would have been only 8.6 percentage points. Similarly, actual service-led growth was 18.2% as opposed to 14.7% under the homothetic counterfactual. The externality from non-homothetic demand amplified overall service transformation and growth by a factor of 1.22-1.24.

Under non-homothetic demand, the model also requires a smaller change in A_M to generate the observed service transformation. In the full estimated model, A_M grew by a population-weighted average of 0.25 log-points across all microregions. If demand was homothetic, it would have needed to grow by an average of 0.31 log-points to match the observed expansion of modern services.

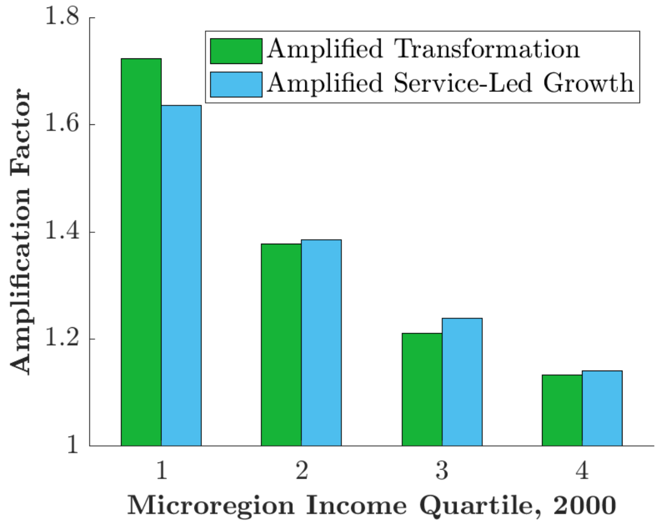
5.2.1 Heterogeneous Amplification

The amplification from non-homothetic demand is significant, but its magnitude varies between poorer and richer microregions. Figure 14 shows how the degree of amplification varies between microregions in each quartile of 2000 aggregate income. To do this, I compute the "amplification factor" in each quartile. This is the ratio of the actual change in quartile-specific outcome Υ_q over 2000-2010 to the counterfactual change if demand was homothetic:

$$\text{Amplification Factor}_q = \frac{\Delta\Upsilon_q|\chi_c > 0}{\Delta\Upsilon_q|\chi_c = 0} \tag{28}$$

I compute the amplification factor for both the service transformation and service-led growth.

Figure 14: Amplification by Microregion Income



Amplification is significantly higher in the poorest quartile of microregions, at a factor of 1.64-1.72, and declines monotonically with income to 1.13-1.14 in the richest quartile. The externality from non-homothetic demand acts as a stronger big push-style amplifying force in poor economies, where labor market frictions are more severe and consumers are more constrained by the costs of modern consumption.

5.3 Frictions Imply Large Reallocation Gains

Changes in labor market search frictions – parameterized by worker search costs κ_w – did not cause significant service transformation from 2000 to 2010, because the underlying change in search costs was modest over this time period. This does not imply that labor market frictions are unimportant. Frictions are substantially higher in the poorest regions of Brazil. They keep the modern service share low and imply that the allocation of labor in these regions is highly distorted.

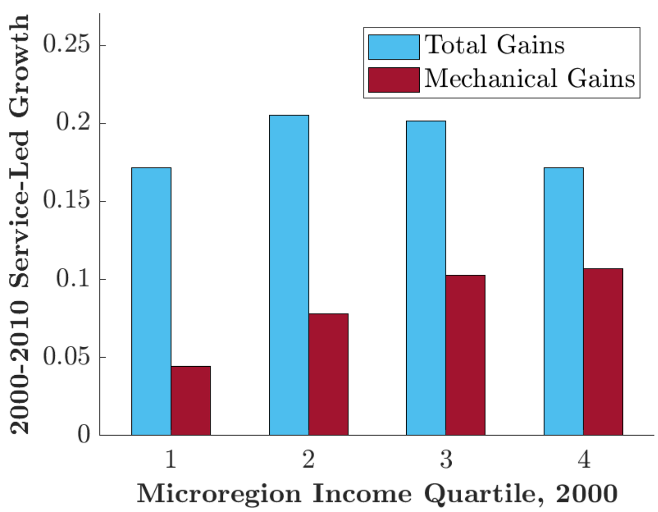
Frictions also imply an opportunity for large gains from reallocation. To illustrate this, I compute 2000-2010 service consumption gains for microregions in each income quartile and separate out the mechanical gains from technological progress. These are gains driven by growth of technology from 2000 to 2010, holding fixed the initial modern expenditure share $\vec{\vartheta}_{M,2000}$. The remaining gains are therefore driven by reallocation and efficiency gains as workers leave behind micro-enterprises like fruit stands and rickshaws, and go to work in modern firms like supermarkets and restaurants.

Recall that total gains can be decomposed:

$$\begin{aligned}\text{Total gains}_q =& \text{Mechanical gain}_q + \text{Efficiency gains}_q \\ \text{Mechanical gain}_q =& \phi \vartheta_{M,q} \Delta \log A_{M,q}\end{aligned} \tag{29}$$

In Figure 15 I present the total and mechanical gains to service consumption by income quartile.[25] While total gains are similar across the income distribution of microregions, the composition of these gains follows a clear pattern. Mechanical gains are small in the poorest microregions. The modern service sector there was initially very small, so the mechanical impact of technological progress in modern services was naturally limited. The more important effect was efficiency gains through reallocation: technological progress pulled workers into the more productive modern sector, where their labor produced a large surplus over the traditional sector. Because frictions caused such a large distortion in the allocation of labor, these poor microregions experienced Lewis-style gains from the recomposition of the service sector. Paired with amplification from demand externalities, reallocation to modern services is a powerful force for service-led growth.

Figure 15: Total and Mechanical Gains in Services by Microregion Income



6 Conclusion

One of the most salient differences between developing and mature economies is the service landscape. Over the development process, modern service firms replace traditional service micro-

[25] As above, I compute the gains individually for each microregion and weight microregions by population within each quartile.

entrepreneurs. This paper uses theory and data to argue that this is more than a superficial change. It is a central driver of service-led growth. The theory is rooted in two basic facts about services: modern service firms are more productive on the margin and in higher demand among rich consumers. As development raises aggregate income, it also raises demand for modern services, pulling additional workers into the more productive modern sector and generating further income gains.

In Brazil, technological progress in modern service firms like supermarkets and chain restaurants catalyzed a process of service-led growth through transformation. Demand effects are strong enough to amplify service-led growth by a factor of 1.2 on average, and by 1.7 in the poorest regions. Developments that improve productivity in service firms, such as expansions of communications infrastructure, are likely to have larger impacts in low-income settings due to feedback effects on demand. A significant share of service-led growth comes from reallocation to more productive jobs. Active labor market policies that help workers transition into these jobs have the potential to generate significant growth in general equilibrium. In contrast, policy stances that implicitly protect traditional service jobs from competition, such as lax enforcement of taxes and labor regulations, have a large aggregate cost.

This paper leaves open several questions that future research could address. For tractability, I use a representative household framework and analyze aggregate consumption gains. If the impact of service transformation on consumers is heterogeneous, so that poor consumers benefit less than rich consumers, then this would explain why policymakers take a permissive stance towards the traditional service sector. I abstract from firm heterogeneity and firm dynamics to focus on workers and consumers, but firm behavior is another important component of service transformation. I also omit human capital accumulation from my analysis, which may contribute to the shift to modern services. Despite these limitations, I believe that my theory of service transformation highlights powerful amplification and reallocation forces that researchers and policymakers should not overlook.

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A Empirical Appendix

A.1 Panel variation: services and development

Figure 16 is the first-difference version of Figure 1a, comparing first differences in log real GDP per capita and the modern share of service employment across country-years. First differences of both variables are residualized against initial log real GDP per capita, initial modern share of service employment, and the time gap between observations in years, since not all countries run their surveys with the same regularity.

Figure 16: Service modernization: first differences

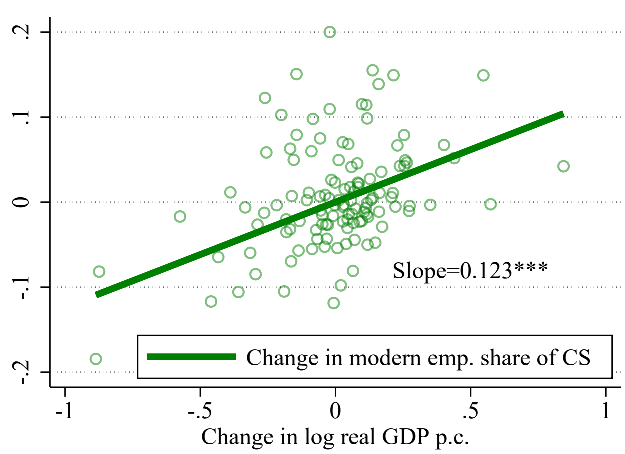
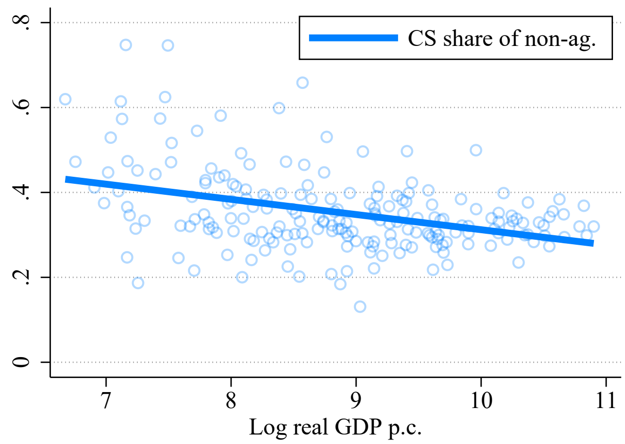


Figure 17 shows that the overall share of consumer services as a percentage of non-agricultural work is relatively flat with respect to development, and in the poorest countries it is just above 40%.

The fact that it represents two-thirds of own-account work in poor countries therefore shows that it is over-represented among own-account work.

Figure 17: Consumer services as a share of non-agricultural work and development



A.2 Estimating the wage premium

To produce Figure 3, I run regressions of the following form among service workers:

$$\log(\text{Earnings}_{ict}) = \beta_0 + \beta_1 \text{Modern Employment}_{ict} + \alpha_{ct} + \vec{\gamma} X_{ict} + u_{ict}, \tag{30}$$

where modern employment is an indicator for wage employment with pension contributions, and α_{ct} is a city-time fixed effect. In all regressions, I use person weights provided by IBGE.

Figure 3 presents different estimates of β_1 for different sub-samples and specifications of controls X_{ict}. The red bars correspond to estimates in the Northeastern cities of Salvador and Recife; the blue bars correspond to estimates in the Southeastern and Southern cities of Sao Paulo, Rio de Janeiro, Belo Horizonte, and Porto Alegre.

The first pair of bars is the "raw" version with only city-time fixed effects; the second pair adds controls for observable characteristics: the worker's sex, race, level of education, age (quadratic), and the industry of the worker's job in that period. The third pair of bars controls for worker individual fixed effects as well as city-time fixed effects and the industry of work. It therefore identifies the effect of modern employment on earnings within worker, among traditional-modern transitioners, after stripping away industry effects. The capped spikes on the bars correspond to robust standard errors of the estimates.

A.3 Estimating the Engel curve

To estimate the Engel curve in Figure 4a, I first construct the modern expenditure share of the household. I use the classification of Bachas et al. (7) and aggregate all consumption expenditures over the previous year. In this classification, the traditional sector consists of non-market consumption sources, vendors withouat a storefront, convenience/corner shops, individual service providers, and informal entertainers. The modern sector consists of specialized shops, large stores, service institutions, and entertainment institutions. The POF has two components: an "expenditure booklet" where households record all expenditures for a week, and a retrospective interview over larger purchases over the previous year. I combine both of these in computing the modern expenditure share, making sure to use the "annualization factor" provided by IBGE: multiplying weekly purchases by 52, monthly purchases by 12, and so on.

The binned scatter of the Engel curve presents the relationship between the modern expenditure share and log income after residualizing both on a battery of controls. These controls are: the number of household residents, the number of earners, the household's age in ten-year dummies, the head's race, the head's gender, the head's education level, and the primary sampling unit (PSU) where the household lives. I provide more detail on PSUs in the next sub-section of this Appendix. In constructing the binned scatter, I use the household weights provided by IBGE.

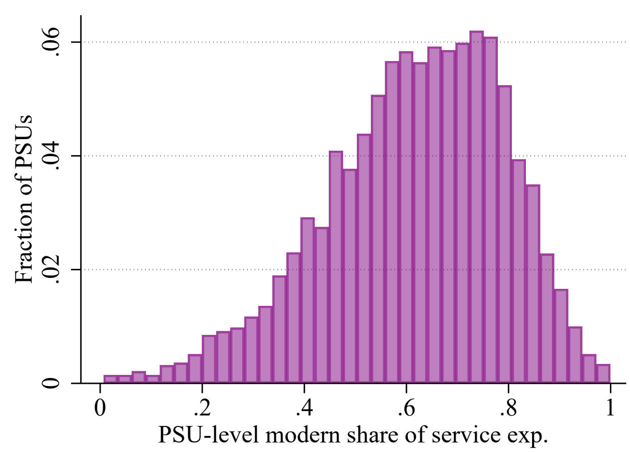
To compute the income gradient that I match in the estimated model, I use a similar procedure but use an instrumental variable for household income. Specifically, I use the set of 3-digit codes for household head's occupation to predict income. I do this to eliminate potential attenuation bias from measurement error on income. The resulting income gradient is a statistically significant 0.14 with a robust standard error of 0.01.

A.4 Location-level modern expenditure shares

One hypothesis for why some households have zero modern expenditures is that there are simply no modern establishments in the local economy. If this were the main reason, it would imply that the modern expenditure share averaged over all local households should be close to zero for some places in Brazil. I test this implication by examining the histogram of modern expenditure shares at the level of primary sampling units (PSUs) of the POF. A PSU is analogous to a census tract (albeit larger in size), and Brazil's national statistical office (IBGE) uses them in two-stage stratified sampling to select the final sample of households in POF. There are 12,800 PSUs in Brazil, and 4,696 of them were randomly sampled for the POF. Within each PSU, 12 to 16 households were selected to be surveyed. In 2008, the average population of a PSU was 15,000, which is equivalent to about 1.5 US ZIP codes. On average, there are two to three PSUs per municipality in Brazil.

Figure 18 presents the histogram of PSU-level average modern expenditure shares. To compute the average PSU-level share, I simply aggregate modern and total expenditure over all households within a given PSU and take their ratio. The PSU-level distribution has a predictable unimodal shape, and notably has very few PSUs with an average modern expenditure share near zero. I therefore conclude that it is unlikely that households consume zero modern services due to having no modern service establishments in the local economy.

Figure 18: Histogram: PSU-level average modern expenditure share



Note: POF sample is 48,706 households from 4,694 primary sampling units (PSUs). PSU-level modern expenditure share is calculated by averaging over all households in the PSU.

B Theoretical Appendix

B.1 Aggregate Demand

Final producers take as given the prices of intermediate inputs. Respectively, these prices are q_M for modern services, q_T for traditional services, and q_X for goods. The respective prices of the final consumption bundles are therefore:

$$P_{\overline{M}} = q_M^{\phi} q_X^{1-\phi}, \quad P_{\overline{T}} = q_T^{\phi} q_X^{1-\phi} \quad \Longrightarrow \frac{P_{\overline{M}}}{P_{\overline{T}}} = \left(\frac{q_M}{q_T}\right)^{\phi} \tag{31}$$

As a result, the price of the modern consumption bundle relative to the traditional one depends only on the relative price of the services used to produce that bundle.

The Cobb-Douglas structure of final consumption also implies that aggregate demand for intermediate inputs is straightforward, conditional on ϑ_M:

$$q_M Y_M = \phi P_{\overline{M}} \Upsilon_M = \phi \vartheta_M e L, \quad q_T Y_T = \phi P_{\overline{T}} \Upsilon_T = \phi(1-\vartheta_M) e L, \quad q_X X = (1-\phi) e L \tag{32}$$

For example, total expenditure on modern services is services' Cobb-Douglas weight ϕ times total expenditure on the modern bundle, $\vartheta_M e L$.

Returning to Equations (3) and Equations (32), it is clear that demand for H_M relative to H_T depends on the CES-like demand system within services, while demand for H_X relative to H_M and H_T depends on Cobb-Douglas demand over goods and services.

$$\begin{gathered} \frac{A_M H_M}{\psi_T H_T} = \frac{\beta_M}{1-\beta_M} \left(\frac{q_M}{q_T}\right)^{-\hat{\xi}} \left(\frac{\frac{e}{q_M^{\phi}} - \chi_c}{\frac{e}{q_M^{\phi}}}\right)^{\frac{\hat{\xi}-1}{\phi}}, \\ \text{where } 1-\hat{\xi} = \phi(1-\xi) \text{ and } e = q_M A_M H_M + q_T \psi_T H_T + \psi_X H_X, \\ \psi_X H_X = \frac{1-\phi}{\phi}(q_M A_M H_M + q_T A_T H_T) \end{gathered} \tag{33}$$

B.2 Out of steady state mechanics of the labor market and proof of steady state allocation

For the sake of notation, let $\ell_w(z,t)$ denote the density of workers of type z in wage work at time t, and $\ell_o(z,t)$ denote the density in own-account work. It must be that $\ell_w(z,t) + \ell_o(z,t) = f(z)L$, where $f(\cdot)$ is the density function of the distribution of z. Similarly, let $h_w(z,t)$ and $h_o(z,t)$ denote the measure of efficiency units supplied by type z to wage and own-account work. Then the system of z-specific labor and efficiency unit supply evolves as follows:

$$
\begin{aligned}
d\ell_w(z,t) &= \lambda(\theta(z,t))\ell_o(z,t)s(z,t) - \delta\ell_w(z,t) \\
\ell_w(z,0) &= 0 \ \forall z \\
s(z,t) &= \mathbf{1}\{z \le z_w(t)\} \\
h_o(z,t) &= \ell_o(z,t)z\mathbb{E}[h|z] \\
h_w(z,t) &= (\ell_w(z,t) - \kappa_v\theta(z,t)\ell_o(z,t)s(z,t))\mathbb{E}[h|z] \\
&,
\end{aligned}
\tag{34}
$$

where $s(z,t)$ is an indicator for whether type z searches at time t, and $\mathbb{E}[h|z]$ is the expected absolute ability of a type-z worker based on the joint distribution of z and h. Note that efficiency units of wage labor must take into account those efficiency units used for vacancy-posting.

The aggregate instantaneous supply of wage and own-account labor then integrates over all z. However, aggregate supply of efficiency units must account for efficiency units lost to the "entry cost" $\mathcal{K}_w$ of search, which depends on the entry flow $\mathcal{E}_t$ of efficiency units into search at time t. Aggregate revenue multiplies efficiency unit supplies by their respective skill prices.

$$
\begin{aligned}
\text{Own-account labor:} \quad & L_o(t)dt = \int_0^\infty \ell_o(z,t)dzdt \\
\text{Own-account efficiency units:} \quad & H_o(t)dt = \int_0^\infty h_o(z,t)dzdt \\
\text{Wage labor:} \quad & L_w(t)dt = \int_0^\infty \ell_w(z,t)dzdt \\
\text{Wage efficiency units:} \quad & H_w(t)dt = \int_0^\infty h_w(z,t)dzdt - \mathcal{K}_w\mathcal{E}_t, \\
\text{Entry flow:} \quad & \mathcal{E}_t = f(z_w(t))\mathbb{E}[h|z_w(t)]dz_w(t) \\
\text{Aggregate revenue:} \quad & R(t)dt = \pi_w(t)H_w(t)dt + \pi_o(t)H_o(t)dt
\end{aligned}
\tag{35}
$$

The entry flow depends on the density of workers near the extensive margin of search $f(z_w(t))$, their average absolute ability, and the speed $dz_w(t)$ at which the margin is changing.

In steady state, $d\ell(z,t) = 0$ for all z and $dz_w(t) = 0$. Imposing these conditions on the system above leads to the steady state allocation of the labor market in partial equilibrium.

B.2.1 Steady State Equilibrium Supply

Steady state equilibrium supply can in fact also be written as a function of the relative service price $\frac{q_M}{q_T}$. To see this, substitute $\frac{w_f}{w_o}=\frac{q_M A_M}{q_T A_T}$ into the equations for labor market tightness $\theta(z)$ and worker sorting z_w:

$$\begin{aligned} \kappa_v \theta(z) &= (1-\eta)\lambda(\theta(z))\frac{1-\frac{q_T A_T}{q_M A_M}z}{\rho+\delta+\eta\lambda(\theta(z))} \\ z_w &= \frac{q_M A_M}{q_T A_T}\left(1-\frac{\rho+\delta+\eta\lambda(\theta(z_w))}{\eta\lambda(\theta(z_w))}\kappa_w\right) \end{aligned} \tag{36}$$

And with some steady state accounting, it is possible to write the aggregate supply of wage and own-account efficiency units in terms of z_w and the function $\theta(z)$:

$$\begin{aligned} H_T = H_o &= \int_0^{z_w} \frac{\delta}{\delta+\lambda(\theta(z))} z\mathbb{E}[h|z]f(z)dz + \int_{z_w}^{\infty} z\mathbb{E}[h|z]f(z)dz \\ H_M + H_X = H_w &= \int_0^{z_w} \frac{\lambda(\theta(z))-\delta\kappa_v\theta(z)}{\delta+\lambda(\theta(z))}\mathbb{E}[h|z]f(z)dz, \end{aligned} \tag{37}$$

where H_o is the total supply of efficiency units to own-account work, and H_w is the total supply to wage work.

B.3 Proof of Proposition 1

Market clearing conditions are as follows:

Product Market Clearing:

$$\begin{aligned} \phi\vartheta_{M,r} e_r L_r &= q_{M,r} Y_{M,r} = q_{M,r} A_{M,r} H_{M,r} \\ \phi(1-\vartheta_{M,r}) e_r L_r &= q_{T,r} Y_{T,r} = q_{T,r} A_{T,r} H_{T,r} \\ (1-\phi) e_r L_r &= X_{W,r} = \mathbf{A}_{X,r} H_{X,r} \end{aligned} \tag{38}$$

Labor Market Clearing:

$$\begin{aligned} H_{M,r} + H_{X,r} &= \overline{H}_{w,r} \\ H_{T,r} &= \overline{H}_{o,r} \end{aligned} \tag{39}$$

First, use the product market clearing conditions and the fact that $\pi_{w,r} = q_{M,r} A_{M,r} = \mathbf{A}_{X,r}$ and $\pi_{o,r} = q_{T,r} A_{T,r}$. By dividing total expenditures on goods and modern services by total expenditures on traditional services, it is possible to define the relative skill price $\frac{\pi_{w,r}}{\pi_{o,r}}(z_{w,r},\vartheta_{M,r})$ recursively:

$$\frac{\pi_{w,r}}{\pi_{o,r}}(z_{w,r},\vartheta_{M,r})=\frac{1-\phi(1-\vartheta_{M,r})}{\phi(1-\vartheta_{M,r})}\frac{\overline{H}_o\left(\frac{\pi_{w,r}}{\pi_{o,r}},z_{w,r}\right)}{\overline{H}_w\left(\frac{\pi_{w,r}}{\pi_{o,r}},z_{w,r}\right)} \tag{40}$$

This also implies that the aggregate supply of efficiency units can be written $\overline{H}_w(z_{w,r}\vartheta_{M,r}),\overline{H}_o(z_{w,r}\vartheta_{M,r})$. It is also straightforward to show that sector-specific efficiency units are functions of $(z_{w,r},\vartheta_{M,r})$. For example:

$$H_M(z_{w,r},\vartheta_{M,r})=\frac{\phi\vartheta_{M,r}}{1-\phi(1-\vartheta_{M,r})}\overline{H}_w(z_{w,r},\vartheta_{M,r}) \tag{41}$$

Next, relative expenditures on modern and traditional services imply the relative service price $\frac{q_{M,r}}{q_{T,r}}(z_{w,r},\vartheta_{M,r})$:

$$\frac{q_{M,r}}{q_{T,r}}(z_{w,r},\vartheta_{M,r})=\frac{\vartheta_{M,r}}{1-\vartheta_{M,r}}\frac{A_{T,r}}{A_{M,r}}\frac{H_T(z_{w,r},\vartheta_{M,r})}{H_M(z_{w,r},\vartheta_{M,r})} \tag{42}$$

Real modern purchasing power $\frac{e_r}{P_{\overline{M}}}(z_{w,r},\vartheta_{M,r})$ can also be found by manipulating product market clearing conditions:

$$\frac{e}{P_{\overline{M}}}(z_{w,r},\vartheta_{M,r})=\frac{1}{L_r}\left(\frac{A_{M,r}H_M(z_{w,r},\vartheta_{M,r})}{\phi\vartheta_{M,r}}\right)^{\phi}\left(\frac{\mathbf{A}_{X,r}H_X(z_{w,r},\vartheta_{M,r})}{1-\phi}\right)^{1-\phi} \tag{43}$$

Next, recall the expression for $z_{w,r}$:

$$z_{w,r}=\frac{\pi_{w,r}}{\pi_{o,r}}\left(1-\frac{\rho+\delta+\eta\lambda(\theta(z_{w,r}))}{\eta\lambda(\theta(z_w))}\rho\mathcal{K}_{w,r}\right) \tag{44}$$

The matching rate $\lambda(\theta(z_{w,r}))$ is a function only of $z_{w,r}$, $\frac{\pi_{w,r}}{\pi_{o,r}}$, and exogenous parameters. $z_{w,r}$ therefore depends only on the relative skill price $\frac{\pi_{w,r}}{\pi_{o,r}}$. And I have already established that $\frac{\pi_w}{\pi_o}(z_{w,r},\vartheta_{M,r})$ is itself a function of $(z_{w,r},\vartheta_{M,r})$. The equation above is therefore a recursive definition of the function $z_{w,r}=z_w^*(\vartheta_{M,r})$.

Similarly, the expression pinning down $\vartheta_{M,r}$ depends on the relative service price $\frac{q_{M,r}}{q_{T,r}}$ and real modern purchasing power $\frac{e}{P_{\overline{M},r}}$:

$$\frac{\vartheta_{M,r}}{1-\vartheta_{M,r}}=\frac{\beta_M}{1-\beta_M}\left(\frac{q_{M,r}}{q_{T,r}}\right)^{1-\xi}\left(\frac{\frac{e_r}{P_{\overline{M},r}}}{\frac{e_r}{P_{\overline{M},r}}-\chi_c}\right)^{\frac{1-\xi}{\phi}}\mathbf{1}\left\{\frac{e_r}{P_{\overline{M},r}}>\chi_c\right\} \tag{45}$$

We have seen that these are functions $\frac{q_{M,r}}{q_{T,r}}(z_{w,r},\vartheta_{M,r})$ and $\frac{e}{P_M}(z_{w,r},\vartheta_{M,r})$. The equation above therefore defines $\vartheta_M^*(z_{w,r})$ recursively.

B.4 Proof of Comparative Statics of H_M

By Proposition 1, we know that H_M can be written as $H_M(z_w,\vartheta_M)$. Therefore we can decompose the comparative static:

$$\frac{dH_M}{d\omega}=\frac{\partial H_M}{\partial z_w}\frac{dz_w}{d\omega}+\frac{\partial H_M}{\partial \vartheta_M}\frac{d\vartheta_M}{d\omega} \tag{46}$$

Next we need to find the total derivative of $z_w = z_w^*(\vartheta_M|\vec{\omega})$ with respect to ω, and likewise for $\vartheta_M = \vartheta_M^*(z_w|\vec{\omega})$. This gives us a system in terms of partial derivatives:

$$\begin{aligned}\frac{dz_w}{d\omega}&=\frac{\partial z_w^*}{\partial\omega}+\frac{\partial z_w^*}{\partial\vartheta_M}\frac{d\vartheta_M}{d\omega}\\ \frac{d\vartheta_M}{d\omega}&=\frac{\partial\vartheta_M^*}{\partial\omega}+\frac{\partial\vartheta_M^*}{\partial z_w}\frac{dz_w}{d\omega}\end{aligned} \tag{47}$$

Solving this system allows us to write the total derivatives in terms of partial derivatives:

$$\begin{aligned}\frac{dz_w}{d\omega}&=\frac{1}{1-\frac{\partial z_w^*}{\partial\vartheta_M}\frac{\partial\vartheta_M^*}{\partial z_w}}\left(\frac{\partial z_w^*}{\partial\omega}+\frac{\partial z_w^*}{\partial\vartheta_M}\frac{\partial\vartheta_M^*}{\partial\omega}\right)\\ \frac{d\vartheta_M}{d\omega}&=\frac{1}{1-\frac{\partial z_w^*}{\partial\vartheta_M}\frac{\partial\vartheta_M^*}{\partial z_w}}\left(\frac{\partial\vartheta_M^*}{\partial\omega}+\frac{\partial\vartheta_M^*}{\partial z_w}\frac{\partial z_w^*}{\partial\omega}\right)\end{aligned} \tag{48}$$

Plugging these total derivatives back into the decomposed comparative static above gives the required result.

B.5 Discussion: The Up-Front Cost of Job Search

A key part of what allows for large reallocation gains and a potential "big push" is the up-front cost of job search, $\mathcal{K}_w > 0$. It resembles the intertemporal setup of the big push in Murphy et al. (41), where a costly investment today raises aggregate income and demand tomorrow. In my model, the marginal worker is indifferent about search in a PDV sense, but shifting her into job search raises her income and the aggregate value of output $\overline{R}$ in the long run, holding skill prices fixed. If increasing $\overline{R}$ also pushes up the long-run relative skill price $\frac{\overline{\pi}_w}{\overline{\pi}_o}$, an effect I will micro-found in my model of consumer demand, then the marginal worker's search can induce even more workers to search in the long run.

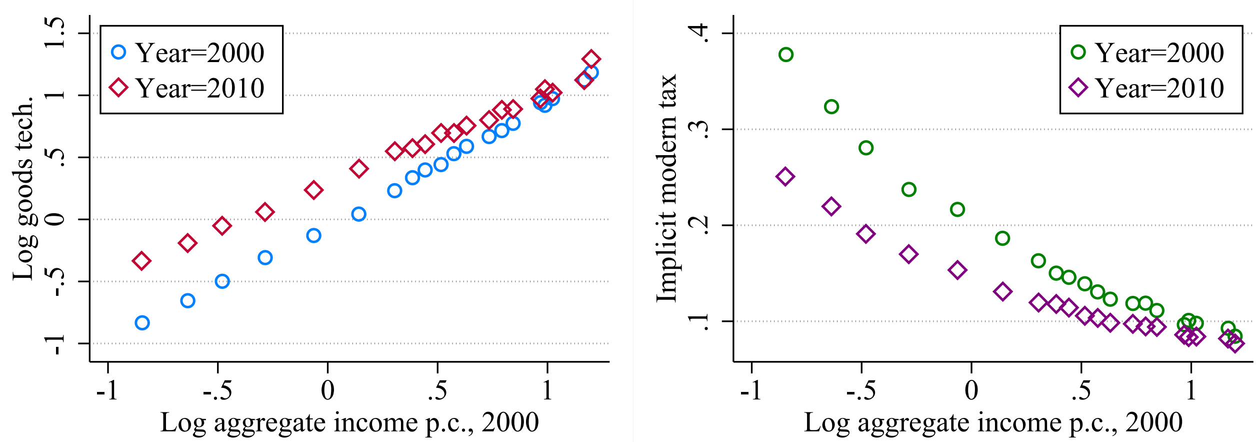
Assuming that all worker search costs are paid up-front is particularly stark, but it is not the only way to get a big push. In fact, the only feature necessary for big push forces is that the worker's returns to search are back-loaded. If they are, then marginal search strictly increases long-run revenue, $\frac{\partial \overline{R}}{\partial z_w} > 0$. The model would have the same feature if, for example, the worker paid a flow cost of search, or if she paid the search entry cost every time she separated from a firm. The advantage of an up-front cost of search is that it affects aggregate income and wages in steady state only through its effect on search decisions, rather than directly entering the resource constraint and wage bargaining.

Do up-front costs generate back-loaded returns to job search? Evidence from recent interventions in developing labor markets suggests that they do.[26] For example, workshops or subsidies for job applications generate persistent earnings and employment gains (1; 2). The positive effects of short-duration vocational training with skill certification are also consistent with this model of the labor market (4).[27]

Search frictions are far from the only micro-foundation of the labor market that could generate a big push when paired with non-homothetic demand. All that is necessary is that moving the marginal worker into wage employment increases the value of aggregate output in the long run.[28] An obvious example is if wage employment must pay a compensating differential for disamenities, as in the "factory wage premium" micro-foundation of Murphy et al. (41). Other suitable micro-foundations include distortions like differential tax enforcement due to informality (56; 17), efficiency wages (33), worker monopoly (44), or employer labor market power (24; 9; 5). The micro-foundation of search with sorting places the model within the large development literature on worker flows (20), own-account work as a response to search frictions (42; 48), and entrepreneurial own-account work (26; 25). It also connects the model to data on worker employment and earnings dynamics, which helps me estimate the model.

Figure 19: Local tradable goods technology and implicit tax on modern consumption

(a) Tradable goods technology Ψ_X **(b)** Product market distortion $\mathcal{T}_c$



C Structural Estimation Appendix

C.1 Other estimated primitives

C.2 Trade shock

Trade in goods To model a trade shock, I assume that each region r is a small open economy that can trade its local good X_r, at price $q_{X,r}$, for the world consumption good X_w, at price $P_{X,w}$. The local good's price $q_{X,r}$ is determined on the global market and is therefore exogenous to conditions within the region r. I normalize the price of the world good $P_{X,w}$ to 1. As a result, the total measure of world goods used in location r, $X_{w,r}$ is as follows:

$$X_{w,r} = q_{X,r} \Psi_{X,r} H_{X,r} = \hat{\Psi}_{X,r} H_{X,r} \tag{49}$$

With the small open economy setup, it is as if the local economy has a technology to transform local goods labor $H_{X,r}$ into world goods $X_{w,r}$, with technology level $\hat{\Psi}_{X,r}$. This "technology" $\hat{\Psi}_{X,r}$ is the product of local physical productivity $A_{X,r}$ and the value $q_{X,r}$ of the locally-produced good.

In my model, the tariff cuts generate exogenous changes in $\hat{\Psi}_{X,rt}$. Recall that $\hat{\Psi}_{X,rt} = q_{X,rt} A_{X,rt}$,

[26] Anecdotally, the author has heard that high-income labor markets, like the academic job market, also work this way.
[27] In the Brazilian context, an intervention currently in the field by Beam, Dahis, Mello, and Navarro-Sola aims to improve labor market matching by developing soft skills.
[28] In static models, the necessary condition is that the marginal wage worker increases aggregate output in a static sense.

where $q_{X,rt}$ is the relative value of region r's locally-produced good compared to the world good X_w. For the average microregion in Brazil, cutting tariffs improved the region's ability to buy world goods, hence raising $q_{X,rt}$. However, microregions that were exposed to significant import competition saw a decline in the relative value of their local good (a drop in $q_{X,rt}$), generating variation in the shock to $\hat{\Psi}_{X,rt}$.

First stage regression I present the first stage regression of the trade shock IV in Table 3.

Table 3: First Stage of 2SLS IV Regression: Trade Shock

	Log income p.c.
Tariff shock (DK17)	-3.755***
	[-4.577,-2.934]
Observations	389
State FE	Yes
Lag Controls	Yes

Note: 95% CI from robust standard errors in brackets. Log income is in 2000, the first Census year post-trade liberalization. Regression controls for 1991 modern employment share, log income, and modern wage premium. Microregions are weighted by 1991 population. For details on construction of the tariff shock, see Dix-Carneiro and Kovak (18).